

Lecture Series 4: Mixed-effects Modelling CM2018

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L4-1 Mixed-effects modelling

Mixed-effects modelling

ESTIMATION OF VARIANCE AND COVARIANCE COMPONENTS*

C. R. HENDERSON

Cornell University



Charles R Henderson
(MSc in Nutrition, PhD in Breeding) developed mixed effects. He pioneered animal breeding – quantitative methods of genetic evaluation of domestic livestock.

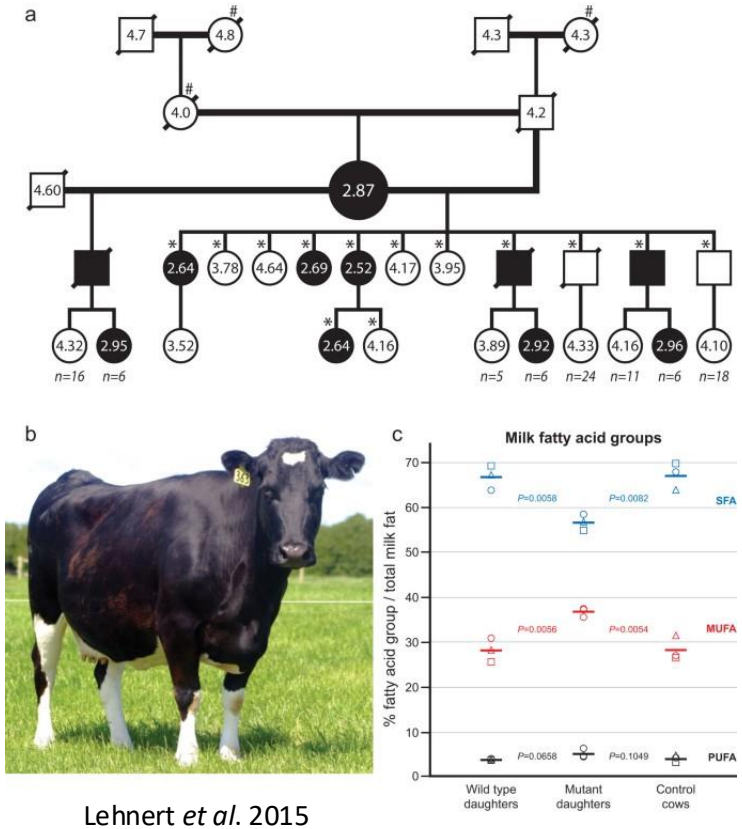
The difficulties in estimating the pertinent variance components are typical of those faced by research workers in animal breeding and in other fields as well. The difficulties in the present example are due to the following causes:

1. Several years' data are involved and time trends are known to be important.
2. The two major classifications of the data are sire and herd. The number of sires exceeds 100 and the number of different herds exceeds 2000.
3. The number of observations per herd-sire subclass varies; the majority being 0.

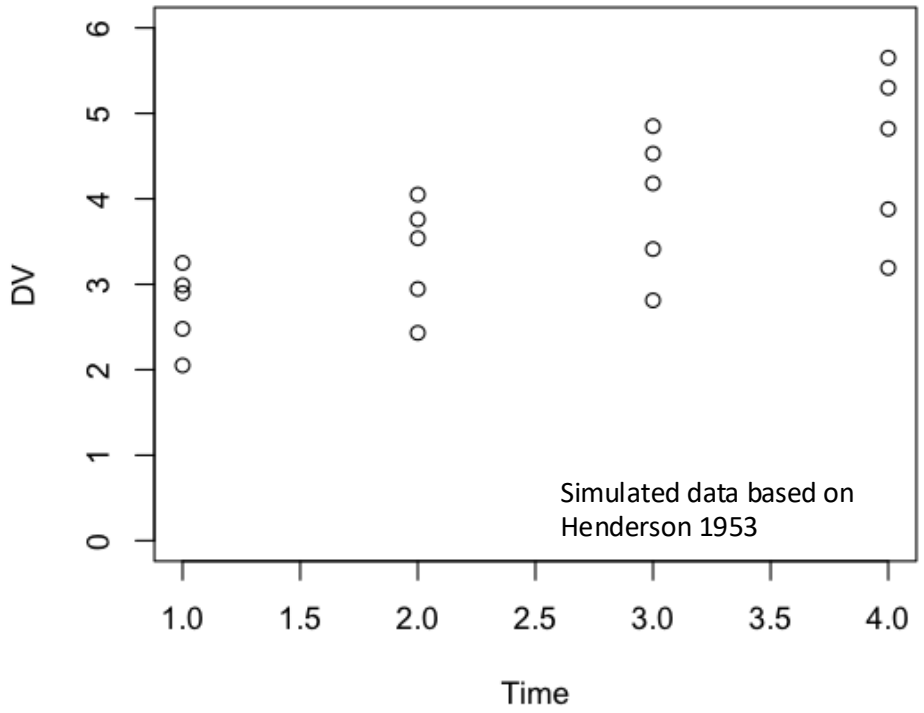
Henderson 1953



Herd data (N herds = 5)



Lehnert *et al*. 2015



ID	Time	DV
1	1	2.1
1	2	2.4
1	3	2.7
1	4	3.2
2	1	2.4
…	…	…

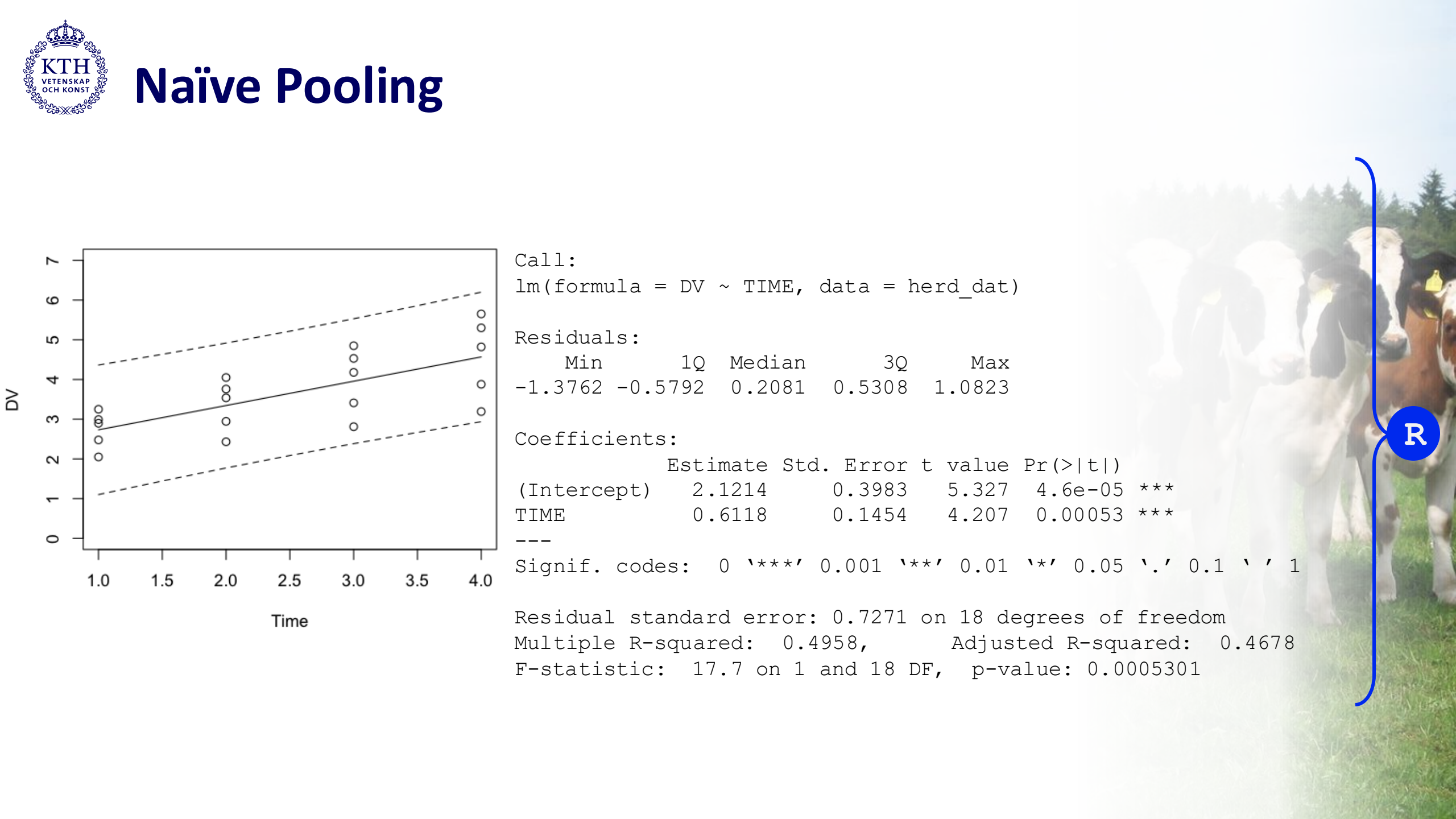
Naïve Pooling


```
Call:
lm(formula = DV ~ TIME, data = herd_dat)

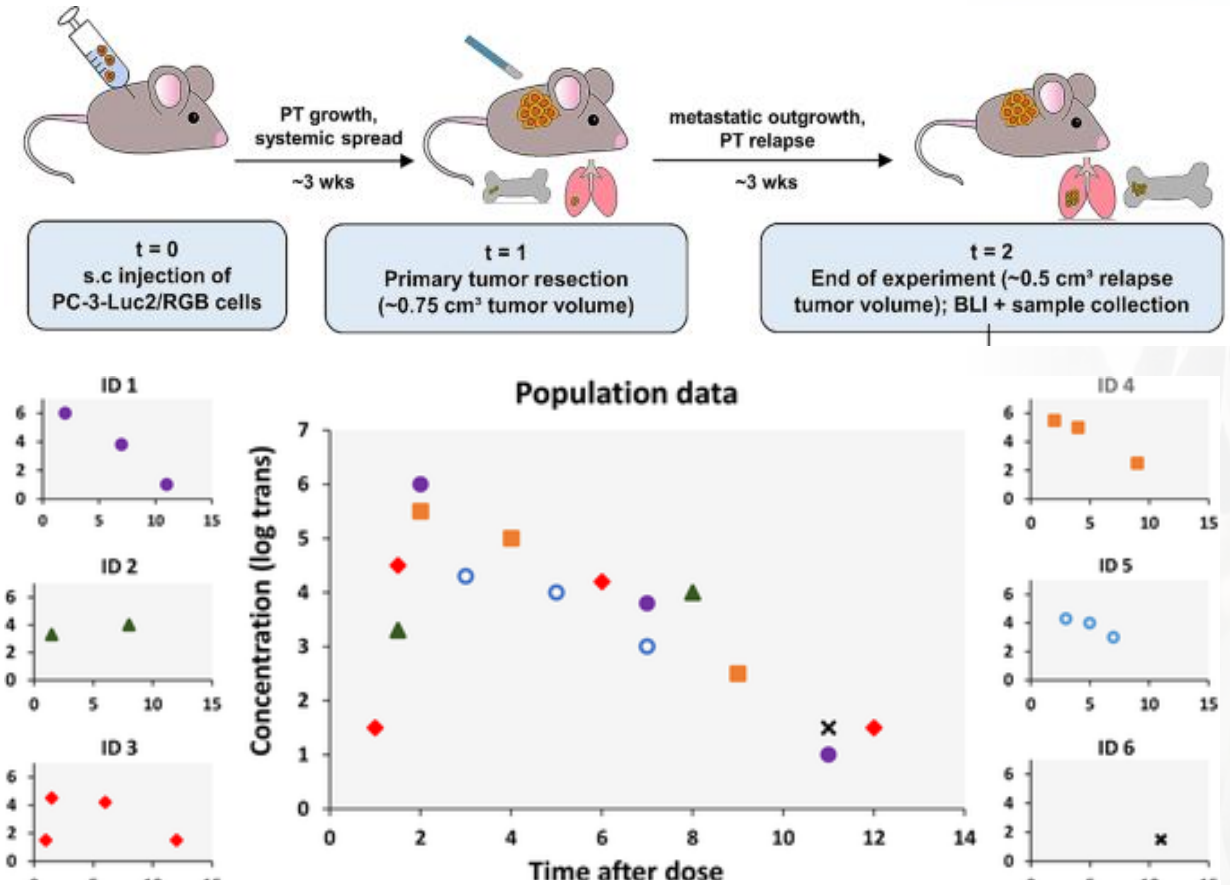
Residuals:
    Min      1Q  Median      3Q     Max
-1.3762 -0.5792  0.2081  0.5308  1.0823

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.1214     0.3983   5.327  4.6e-05 ***
TIME          0.6118     0.1454   4.207  0.00053 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7271 on 18 degrees of freedom
Multiple R-squared:  0.4958,      Adjusted R-squared:  0.4678
F-statistic:  17.7 on 1 and 18 DF,  p-value: 0.0005301
```



Naïve Pooling

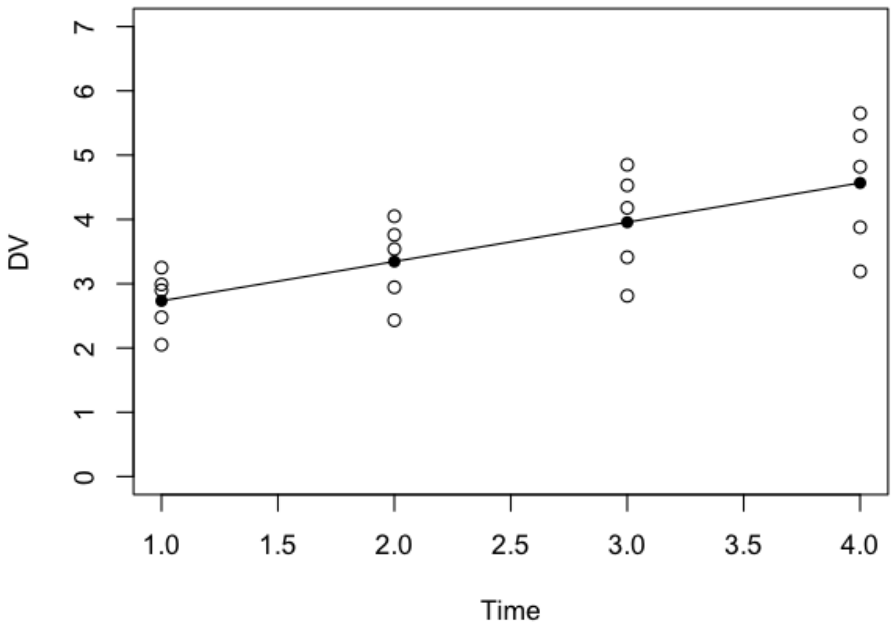


Pharmacokinetics and population pharmacokinetics in pediatric oncology

Sebastiaan D.T. Sassen[1] | C. Michel Zwaan[1,2] | Inge M. van der Sluis[2] | Ron A.A. Mathôt[3]

Sassen *et al*. 2019; Haider *et al*. 2024

Complete Pooling



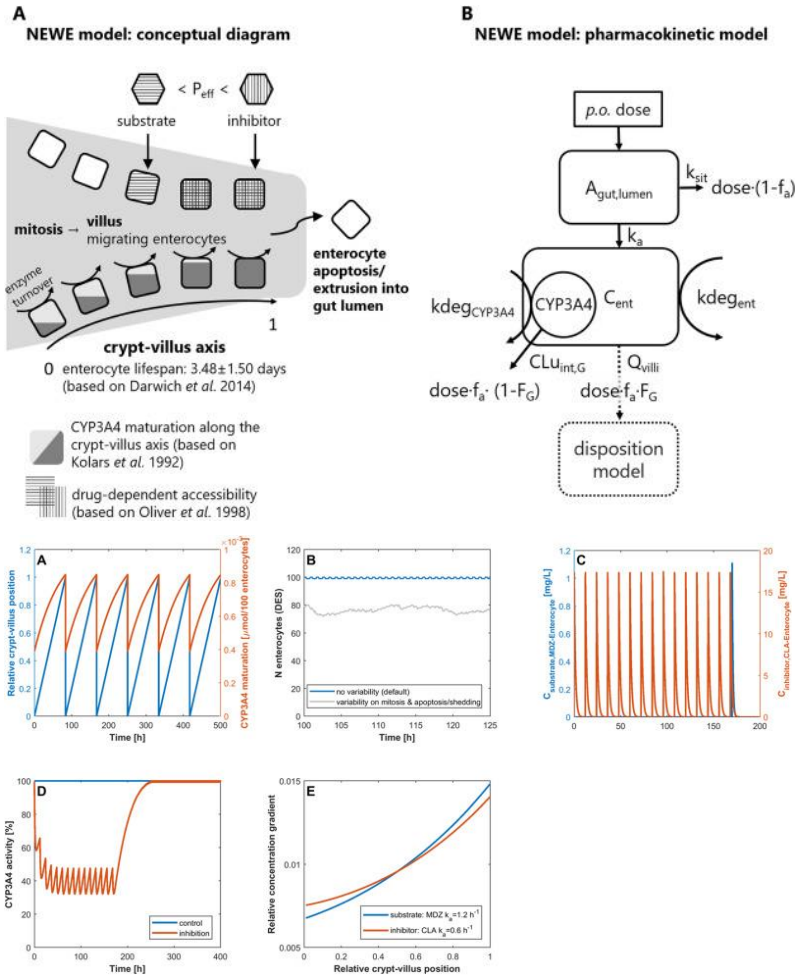

```
Call:
lm(formula = DV ~ TIME, data = c_dat)

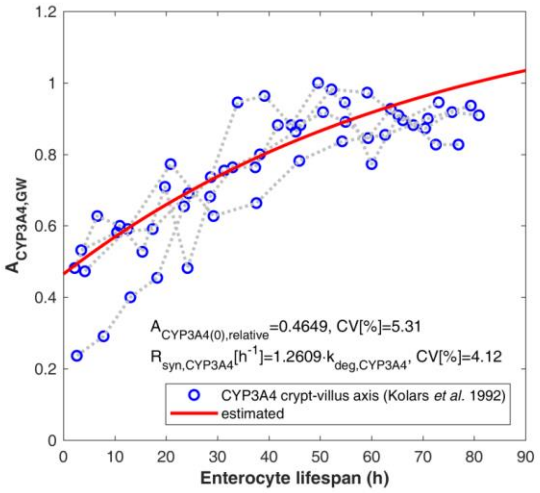
Residuals:
         1          2          3          4
-2.087e-16  3.478e-16 -6.957e-17 -6.957e-17

Coefficients:
             Estimate Std. Error   t value Pr(>|t|)
(Intercept) 2.121e+00  3.615e-16 5.869e+15   <2e-16 ***
TIME        6.118e-01  1.320e-16 4.635e+15   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.951e-16 on 2 degrees of freedom
Multiple R-squared:      1,       Adjusted R-squared:      1
F-statistic: 2.148e+31 on 1 and 2 DF,  p-value: < 2.2e-16
```

Complete Pooling

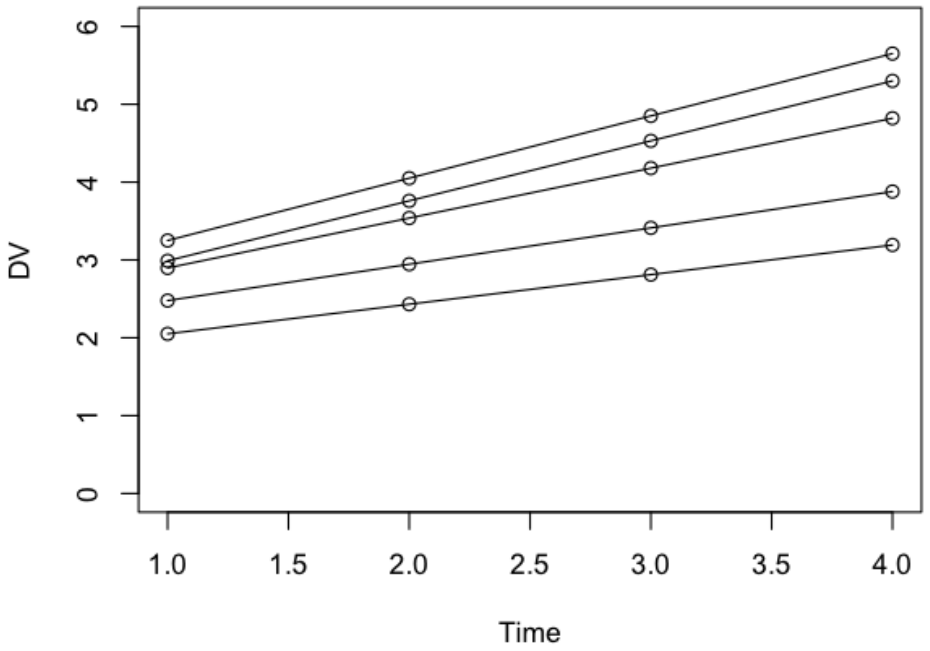






Darwich *et al*. 2019

No Pooling



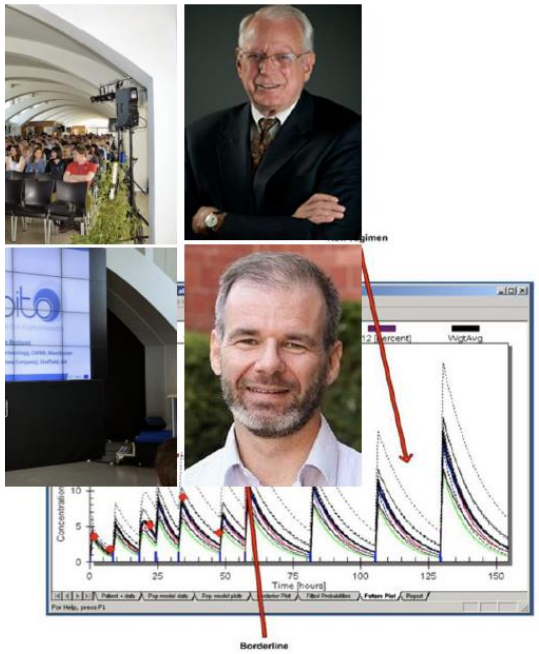

```
Call:
lm(formula = DV ~ TIME + factor(ID) - 1, data = herd_dat)

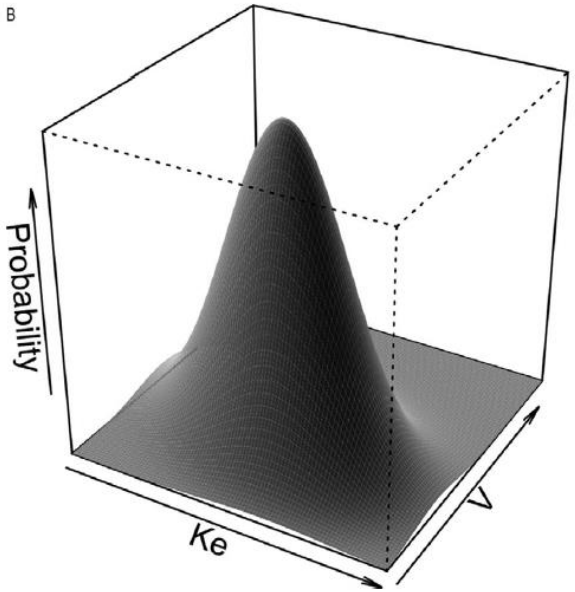
Residuals:
     Min       1Q   Median       3Q      Max
-0.34696 -0.09962  0.00000  0.09962  0.34696

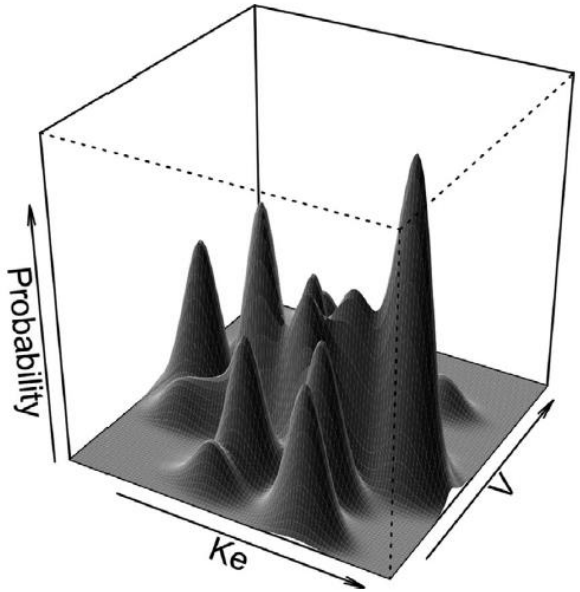
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
TIME         0.61177    0.04408  13.878 1.42e-09 ***
factor(ID)1  1.64905    0.15585  10.581 4.62e-08 ***
factor(ID)2  2.61536    0.15585  16.781 1.14e-10 ***
factor(ID)3  2.92091    0.15585  18.742 2.59e-11 ***
factor(ID)4  1.09210    0.15585   7.007 6.18e-06 ***
factor(ID)5  2.32952    0.15585  14.947 5.33e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2204 on 14 degrees of freedom
Multiple R-squared:  0.9976,      Adjusted R-squared:  0.9966
F-statistic:   977 on 6 and 14 DF,  p-value: < 2.2e-16
```

No Pooling



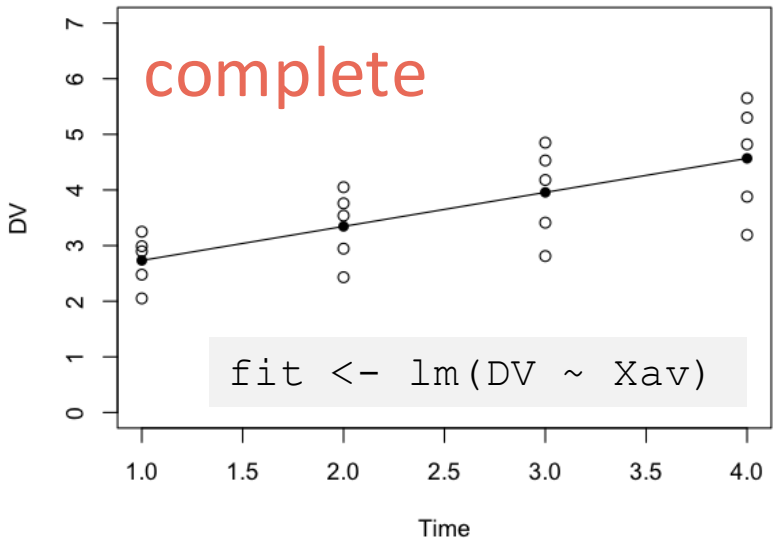


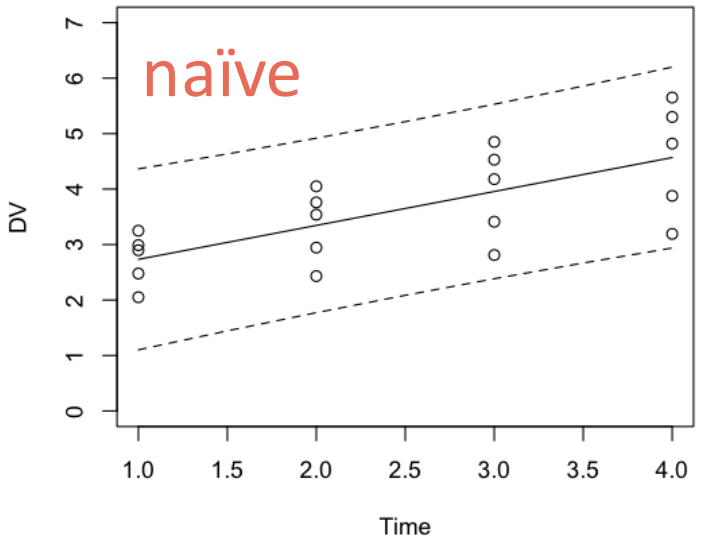


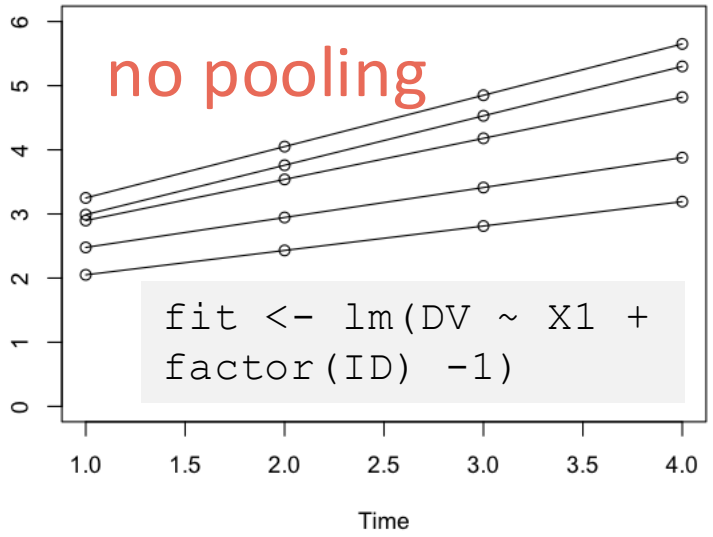
Goutelle *et al*. 2022



Pooling







$\sigma_{\propto} \to 0$

$\sigma_{\propto} \to \infty$

$$\propto_j \sim N(\mu_\alpha, \sigma_{\propto}^2), for\ j = 1, \ldots, J$$

Repeated-measure ANOVA

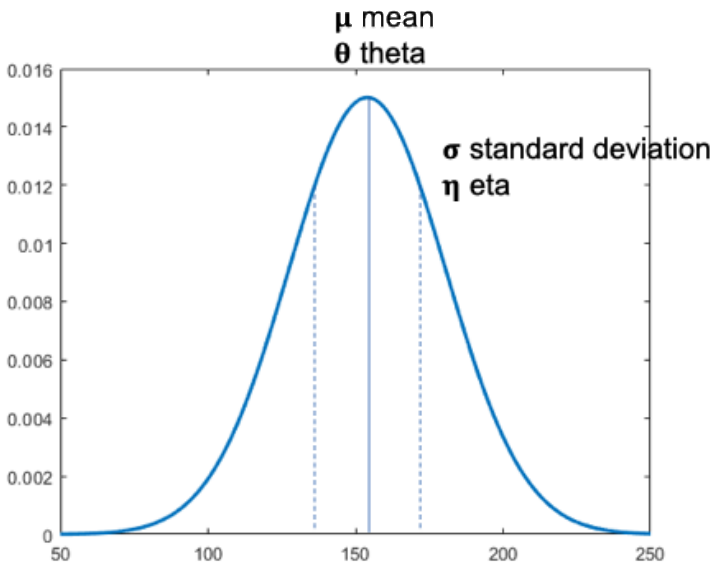
- It handles missing DV data through listwise deletion.
- It treats each response as a different variable.
- Can only account for categorical repeats, not continuous time.

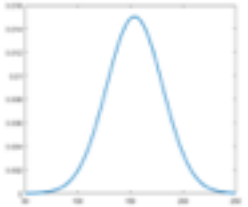


Mixed-effects or partial pooling

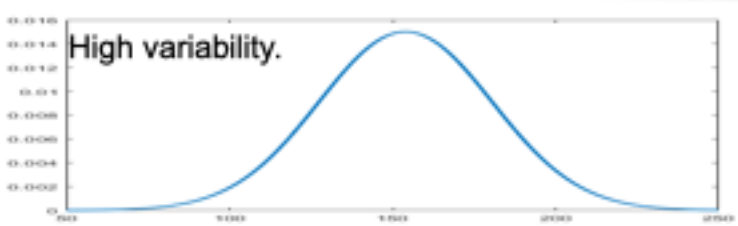
$$\propto_j \sim N(\mu_\alpha, \sigma^2_\propto), for\ j = 1, \ldots, J$$

$$\propto_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\propto^2}} \cdot (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\propto^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\propto^2}} \cdot \mu_\propto$$





Group level standard deviation is small, more pooling. Smoothing of groups with few observations.





Mixed-effects modelling in R

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: DV ~ 1 + TIME + (1 + TIME | ID)
   Data: herd_dat

REML criterion at convergence: -304

Scaled residuals:
       Min         1Q     Median         3Q        Max
-1.620e-06 -4.007e-07  2.642e-08  3.853e-07  1.277e-06

Random effects:
 Groups   Name        Variance  Std.Dev.  Corr
 ID       (Intercept) 3.629e-02 1.905e-01
          TIME        1.255e-02 1.120e-01 0.89
 Residual             2.543e-15 5.043e-08
Number of obs: 20, groups:  ID, 5

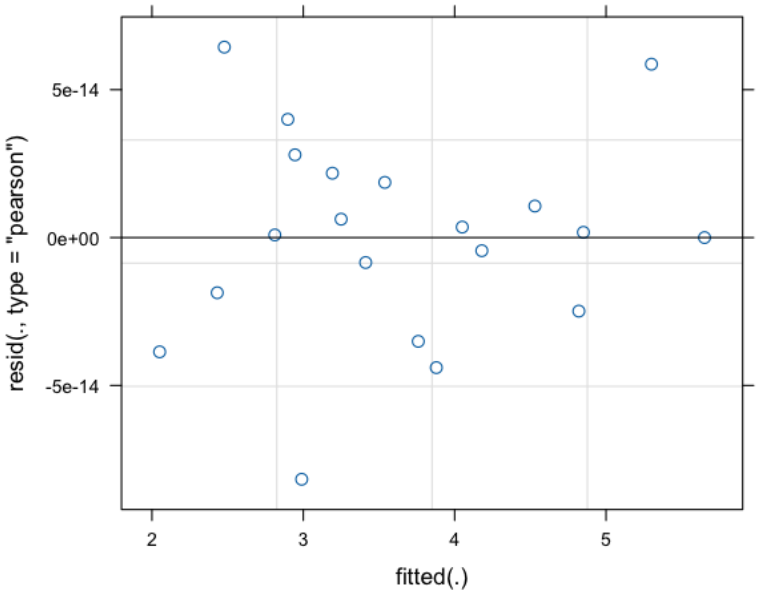
Fixed effects:
            Estimate Std. Error       df t value Pr(>|t|)
(Intercept) 2.118031   0.088702 0.001927   23.88    0.987
TIME        0.607934   0.051939 0.001927   11.71    0.988

Correlation of Fixed Effects:
     (Intr)
TIME 0.900
```

model converged successfully

distribution of residual errors

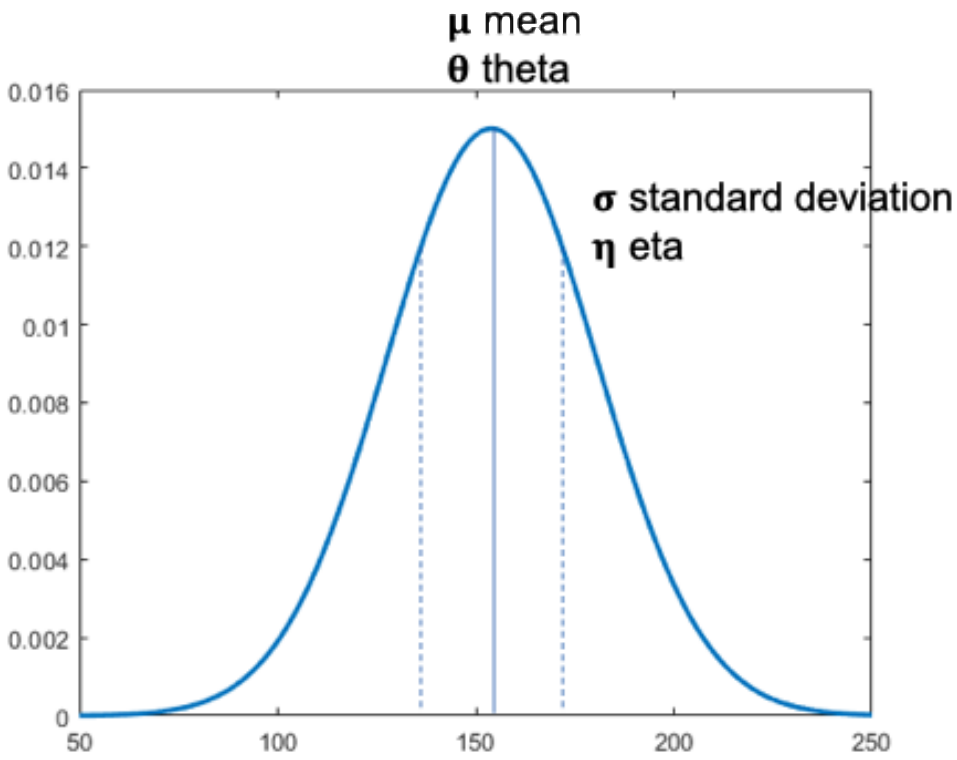
estimates of random effects



- estimates of fixed effects,
- t value,
- df based on Satterthwaite,
- Pr(>|t|) the proportion of the t-distribution at df which is greater than |t value|.

L4-2 Mixed-effects modelling & study design

Mixed-effects combines fixed and random effects

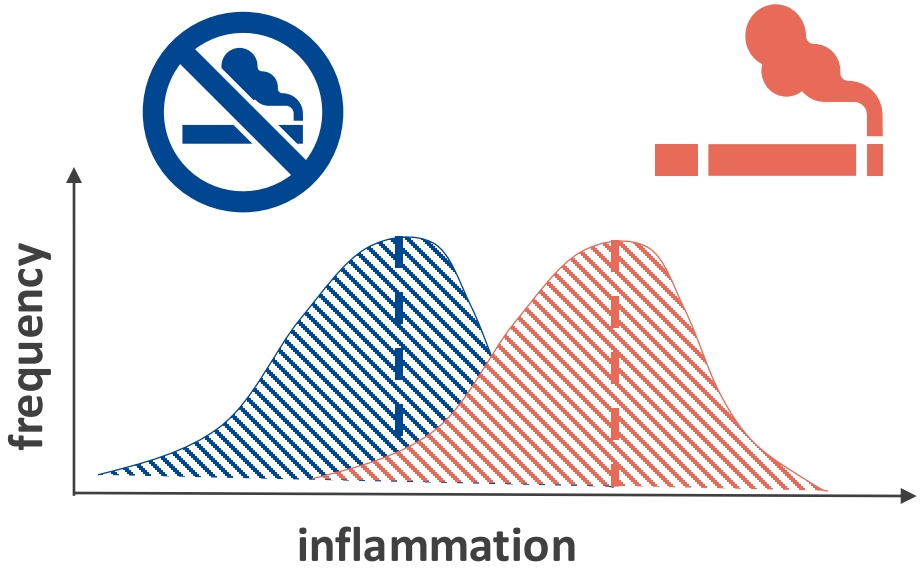


- **Fixed effects** describes the general behaviour in the population, not one particular individual in the sample.
- **Random effects** describe random draws from the population distribution.
- **Mixed-effects models** contain both fixed and random effects.

Lavine 2013, Introduction to Statistical Thought

The effect of smoking on inflammation

H_0: average inflammation does not differ.
H_1: average inflammation differs.



How do we determine if there is a difference?

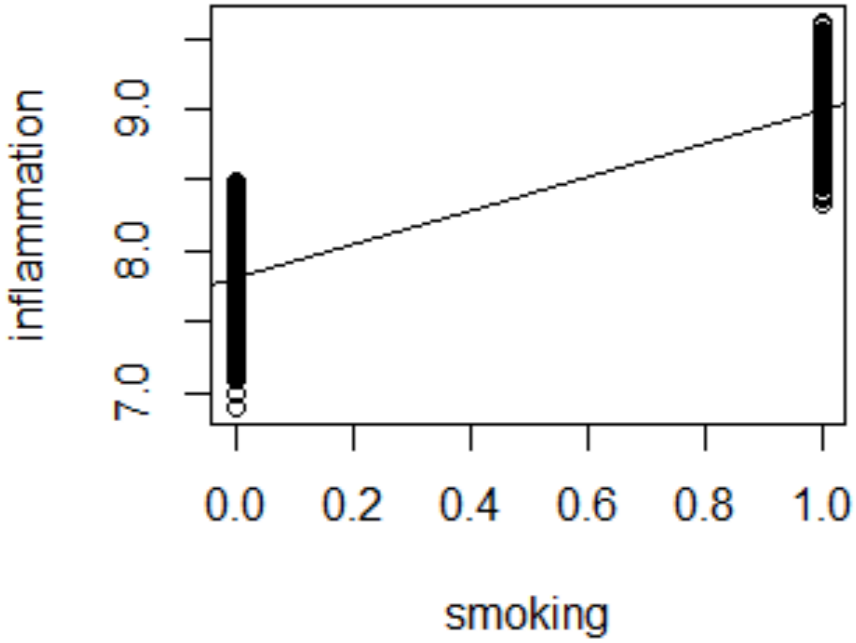
Classical regression model: fixed effects

Let us consider the deterministic formula:

$$\hat{y} = \alpha + x \cdot \beta$$

where y-hat is the predicted, or expected, dependent variable given the predictor, or indicator, β and an intercept of α.

Classical regression model: fixed effects



$$\mathbf{inflammation} = \mathbf{7.8} + \mathbf{1.2} \cdot \boldsymbol{smoking} + \boldsymbol{error}$$

```
fit1 <- lm(inflammation ~ smoking)
summary(fit1)
plot(inflammation ~ smoking)
abline(fit1)
```

Inference & causality

A Frank Statement to Cigarette Smokers

RECENT REPORTS on experiments with mice have given wide publicity to a theory that cigarette smoking is in some way linked with lung cancer in human beings.

Although conducted by doctors of professional standing, these experiments are not regarded as conclusive in the field of cancer research. However, we do not believe that any serious medical research, even though its results are inconclusive should be disregarded or lightly dismissed.

At the same time, we feel it is in the public interest to call attention to the fact that eminent doctors and research scientists have publicly questioned the claimed significance of these experiments.

Distinguished authorities point out:

1. That medical research of recent years indicates many possible causes of lung cancer.
2. That there is no agreement among the authorities regarding what the cause is.
3. That there is no proof that cigarette smoking is one of the causes.
4. That statistics purporting to link cigarette smoking with the disease could apply with equal force to any one of many other aspects of modern life. Indeed the validity of the statistics themselves is questioned by numerous scientists.

We accept an interest in people's health as a basic responsibility, paramount to every other consideration in our business.

We believe the products we make are not injurious to health.

We always have and always will cooperate closely with those whose task it is to safeguard the public health.

For more than 300 years tobacco has given solace, relaxation, and enjoyment to mankind. At one time or another during these years critics have held it responsible for practically every disease of the human body. One by one these charges have been abandoned for lack of evidence.

Regardless of the record of the past, the fact that cigarette smoking today should even be suspected as a cause of a serious disease is a matter of deep concern to us.

Many people have asked us what we are doing to meet the public's concern aroused by the recent reports. Here is the answer:

1. We are pledging aid and assistance to the research effort into all phases of tobacco use and health. This joint financial aid will of course be in addition to what is already being contributed by individual companies.
2. For this purpose we are establishing a joint industry group consisting initially of the undersigned. This group will be known as TOBACCO INDUSTRY RESEARCH COMMITTEE.
3. In charge of the research activities of the Committee will be a scientist of unimpeachable integrity and national repute. In addition there will be an Advisory Board of scientists disinterested in the cigarette industry. A group of distinguished men from medicine, science, and education will be invited to serve on this Board. These scientists will advise the Committee on its research activities.

This statement is being issued because we believe the people are entitled to know where we stand on this matter and what we intend to do about it.

TOBACCO INDUSTRY RESEARCH COMMITTEE

500 FIFTH AVENUE, NEW YORK 36, N. Y.

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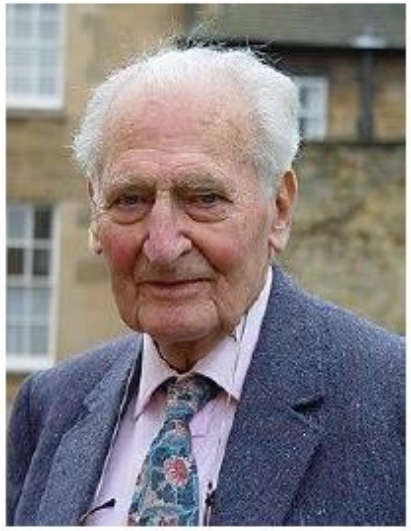
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BRITISH MEDICAL JOURNAL

LONDON SATURDAY SEPTEMBER 30 1950

SMOKING AND CARCINOMA OF THE LUNG

PRELIMINARY REPORT

BY

RICHARD DOLL, M.D., M.R.C.P.

Member of the Statistical Research Unit of the Medical Research Council

AND

A. BRADFORD HILL, Ph.D., D.Sc.

Professor of Medical Statistics, London School of Hygiene and Tropical Medicine; Honorary Director of the Statistical Research Unit of the Medical Research Council

In England and Wales the phenomenal increase in the number of deaths attributed to cancer of the lung provides one of the most striking changes in the pattern of mortality recorded by the Registrar-General. For example, in the quarter of a century between 1922 and 1947 the annual number of deaths recorded increased from 612 to 9,287, or roughly fifteenfold. This remarkable increase is, of course, out of all proportion to the increase of population—both in total and, particularly, in its older age groups. Stocks (1947), using standardized death rates to allow for these population changes, shows the following trend: rate per 100,000 in 1901–20, males 1.1, females 0.7; rate per 100,000 in 1936–9, males 10.6, females 2.5. The rise seems to have been particularly rapid since the end of the first world war; between 1921–30 and 1940–4 the death rate of men at ages 45 and over increased sixfold and of women of the same ages approximately threefold. This increase is still continuing. It has occurred, too, in Switzerland, Denmark, the U.S.A., Canada, and Australia, and has been reported from Turkey and Japan.

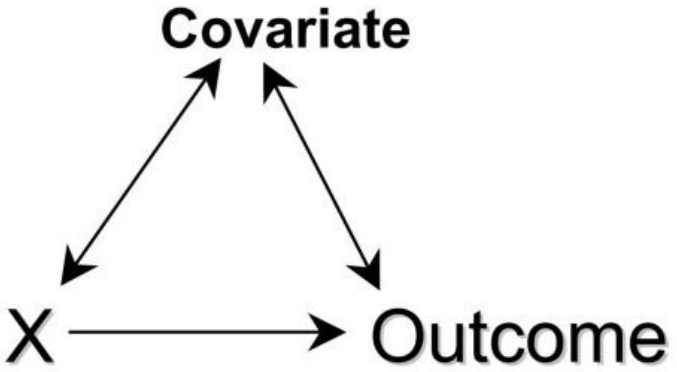
Many writers have studied these changes, considering whether they denote a real increase in the incidence of the disease or are due merely to improved standards of diagnosis. Some believe that the latter factor can be regarded as wholly, or at least mainly, responsible—for example, Willis (1948), Clemmesen and Busk (1947), and Steiner whole explanation, although no one would deny that it may well have been contributory. As a corollary, it is right and proper to seek for other causes.

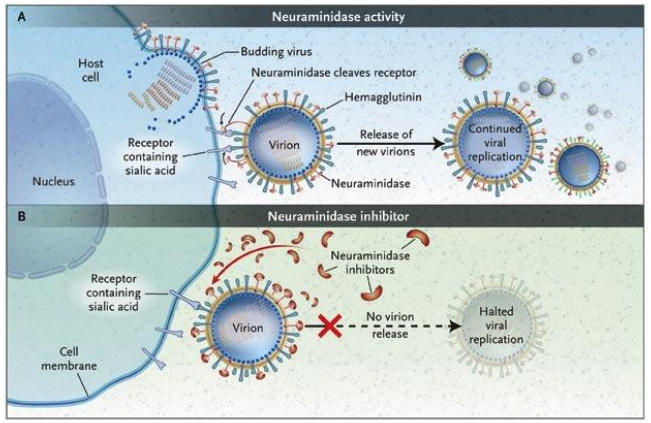
Possible Causes of the Increase

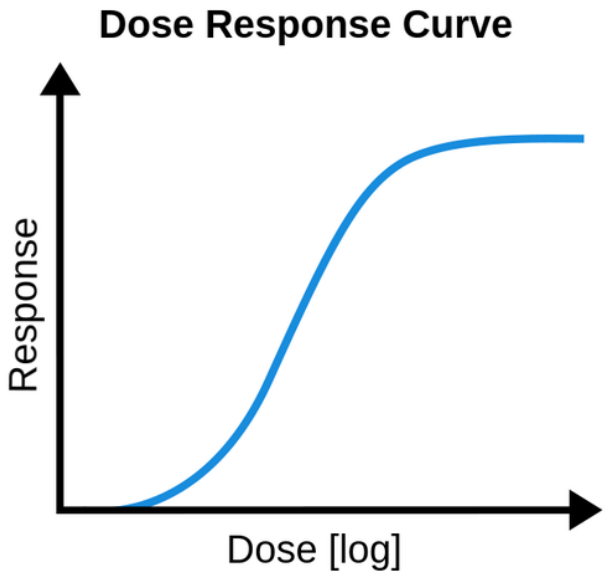
Two main causes have from time to time been put forward: (1) a general atmospheric pollution from the exhaust fumes of cars, from the surface dust of tarred roads, and from gas-works, industrial plants, and coal fires; and (2) the smoking of tobacco. Some characteristics of the former have certainly become more prevalent in the last 50 years, and there is also no doubt that the smoking of cigarettes has greatly increased. Such associated changes in time can, however, be no more than suggestive, and until recently there has been singularly little more direct evidence. That evidence, based upon clinical experience and records, relates mainly to the use of tobacco. For instance, in Germany, Müller (1939) found that only 3 out of 86 male patients with cancer of the lung were non-smokers, while 56 were heavy smokers, and, in contrast, among 86 "healthy men of the same age groups" there were 14 non-smokers and only 31 heavy smokers. Similarly, in America, Schrek and his co-workers (1950) reported that 14.6% of 82 male patients with cancer of the lung were non-smokers, against 23.9% of 522 male patients admitted with cancer of sites other than the upper respiratory and digestive

Bradford Hill criteria

- **Strength of association** effect size and mechanism.
- **Consistency** repeatability and data integration.
- **Specificity** mechanistic understanding of effect.
- **Biological gradient** dose-response curve.
- **Plausability** biological plausability.
- **Coherence** mechanistic studies reinforce epidemiological studies.
- **Temporality** exposure must precede effect.
- **Experiment** response to experimental manipulation.
- **Analogy** similar agents/compounds act similarly.







Fedak, *et al.*, *Emerg Themes Epidemiol (2015) 12:14*

Lady tasting tea




The
Design of Experiments

By
R. A. Fisher, Sc.D., F.R.S.

Oliver and Boyd
Edinburgh: Tweeddale Court
London: 33 Paternoster Row, E.C.
1935



H_0: The lady has no ability to distinguish the tea prepared with milk first from those with milk added later.



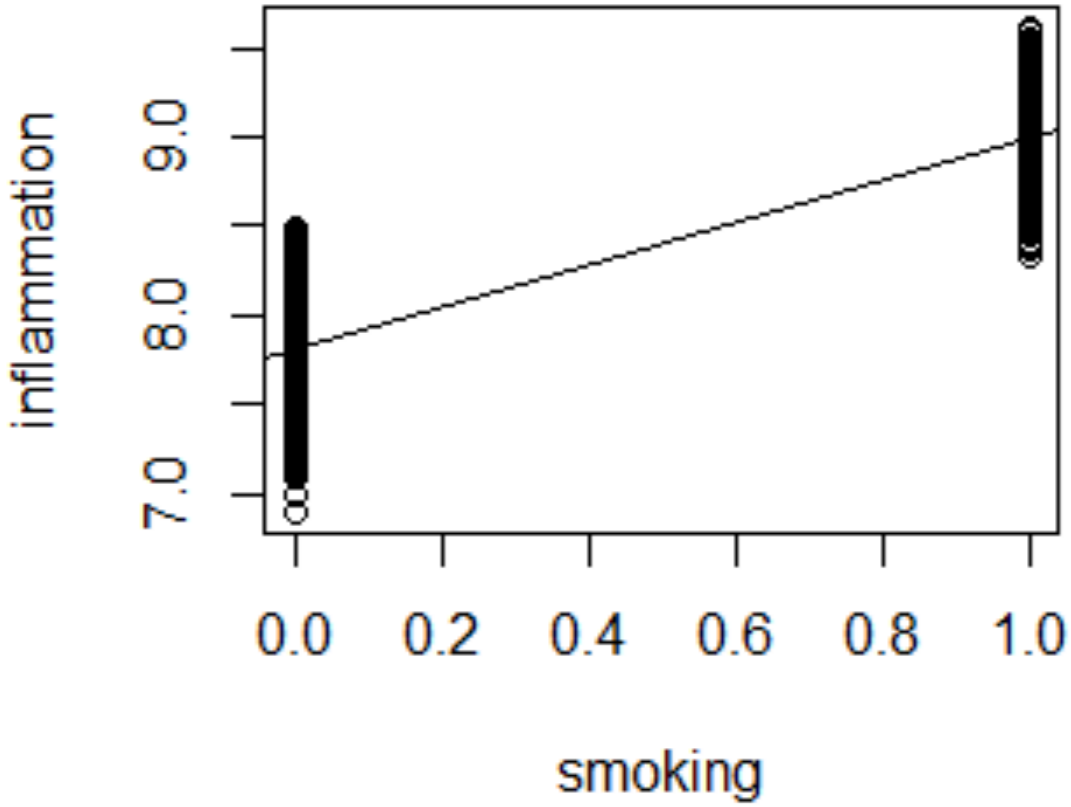
Counting number of successes compared to chance.

Fisher 1935

Rethinking our study design



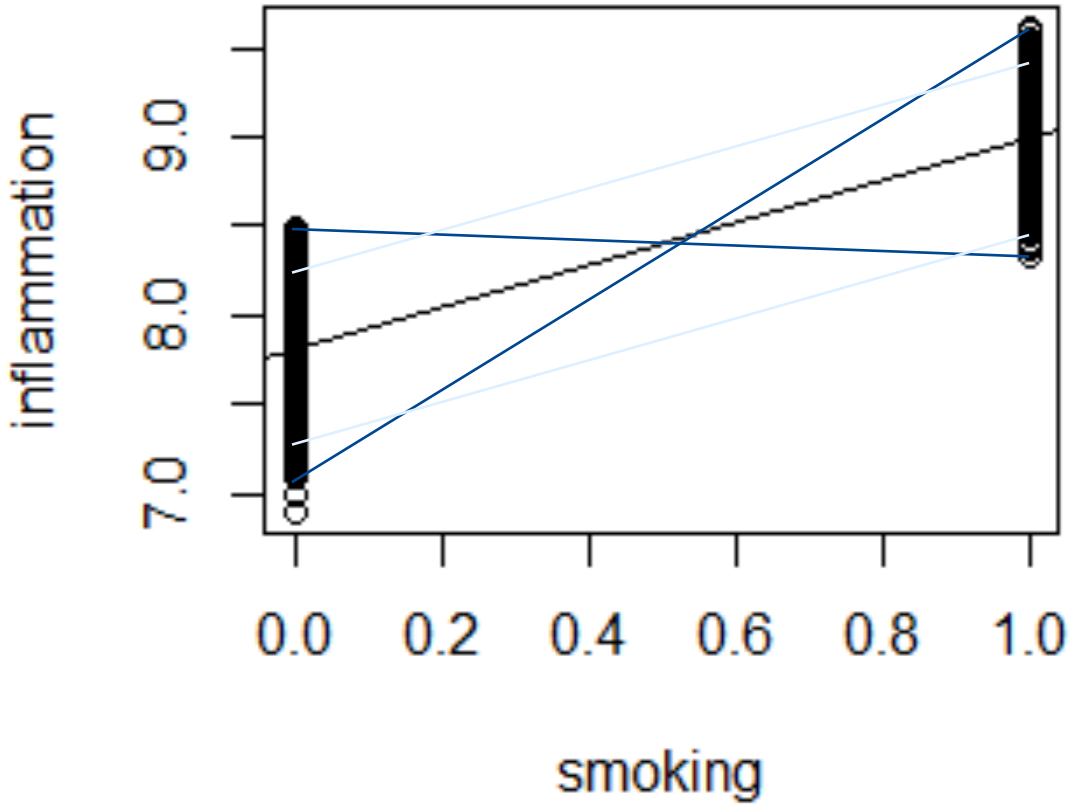
If we formulate this as a follow-up study where we measure inflammation before and after initiation of smoking, in the same individuals.



$$\mathbf{inflammation} = \mathbf{7.8} + \mathbf{1.2} \cdot \boldsymbol{smoking} + \boldsymbol{error}$$

```
fit1 <- lm(inflammation ~ smoking)
summary(fit1)
plot(inflammation ~ smoking)
abline(fit1)
```

If we formulate this as a follow-up study where we measure inflammation before and after initiation of smoking, in the same individuals.



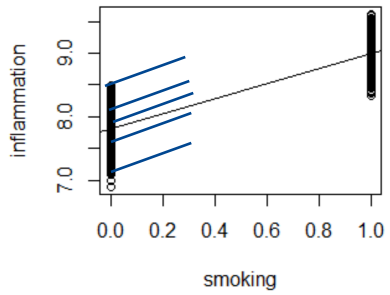
???

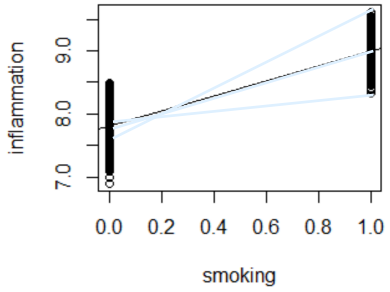
???

$$\mathbf{inflammation} = \mathbf{7.8} + \mathbf{1.2} \cdot \boldsymbol{smoking} + \boldsymbol{error}$$

```
fit1 <- lm(inflammation ~ smoking)
summary(fit1)
plot(inflammation ~ smoking)
abline(fit1)
```

Mixed-effects modelling in a nutshell

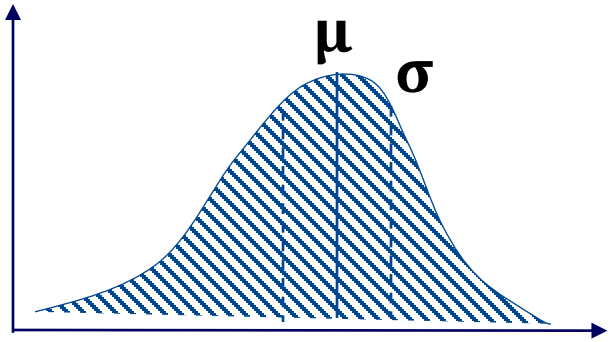




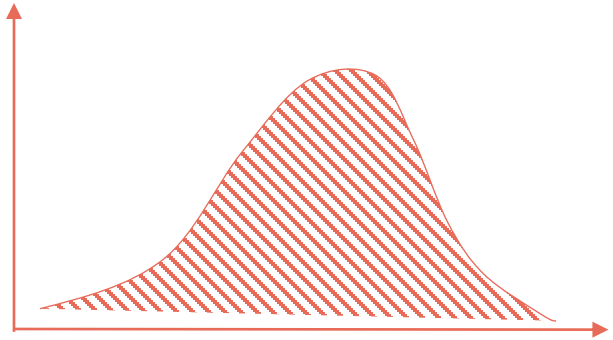
Outcome

$$Inflammation_{Baseline} = \mu \cdot norm(0, \sigma)$$

$$Effect_{Smoking} = \mu \cdot norm(0, \sigma)$$



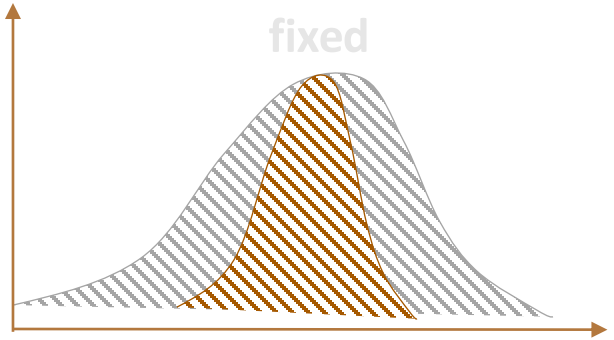
Inflammation at baseline



change in inflammation due to smoking

mixed

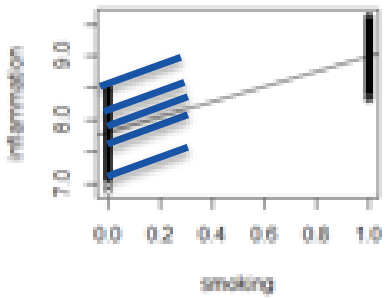
fixed

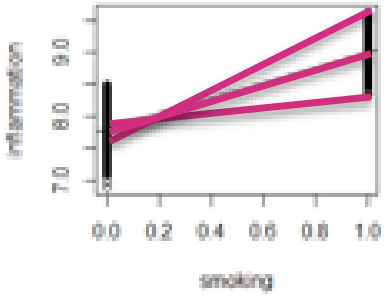


plus some noise!
Residual variability

$$\mathbf{inflammation} = \mathbf{7.8} + \mathbf{1.2} \cdot \boldsymbol{smoking} + \boldsymbol{error}$$

Mixed-effects modelling in a nutshell





Outcome

$$Inflam_{Baseline} = \mu \cdot norm(0, \sigma)$$

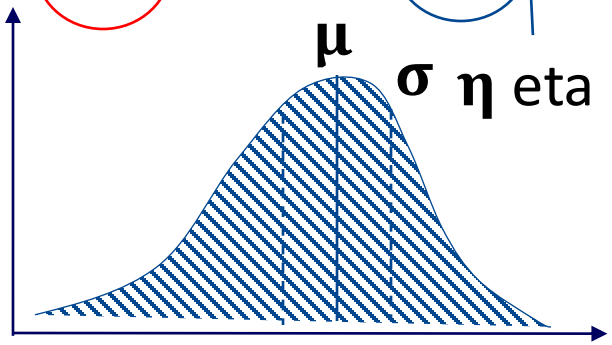
$$Effect_{Smoking} = \mu \cdot norm(0, \sigma)$$

fixed effects
+
random effects

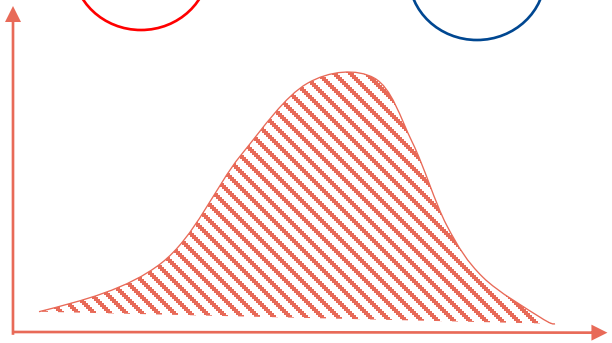
= mixed effects

θ theta

μ **σ** **η** eta



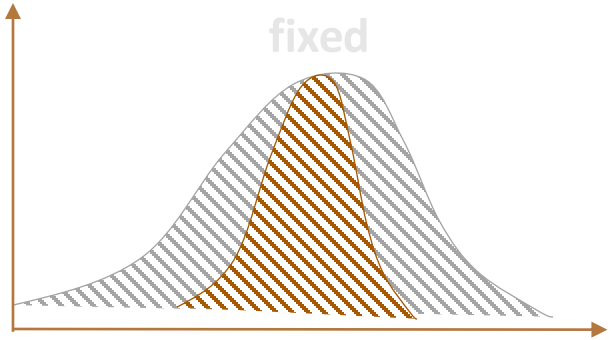
Inflammation at baseline



change in inflammation due to smoking

mixed

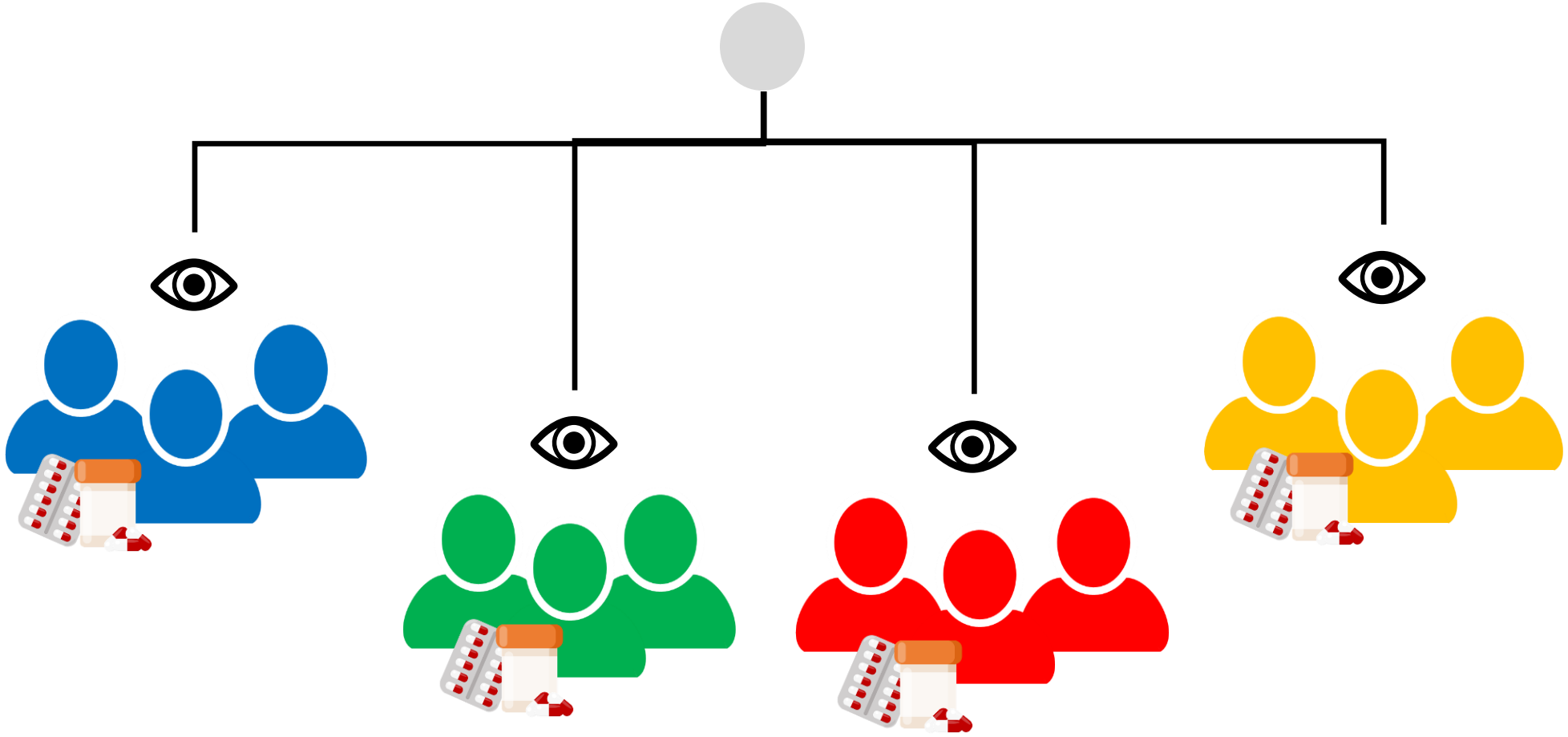
fixed



plus some noise!
Residual variability

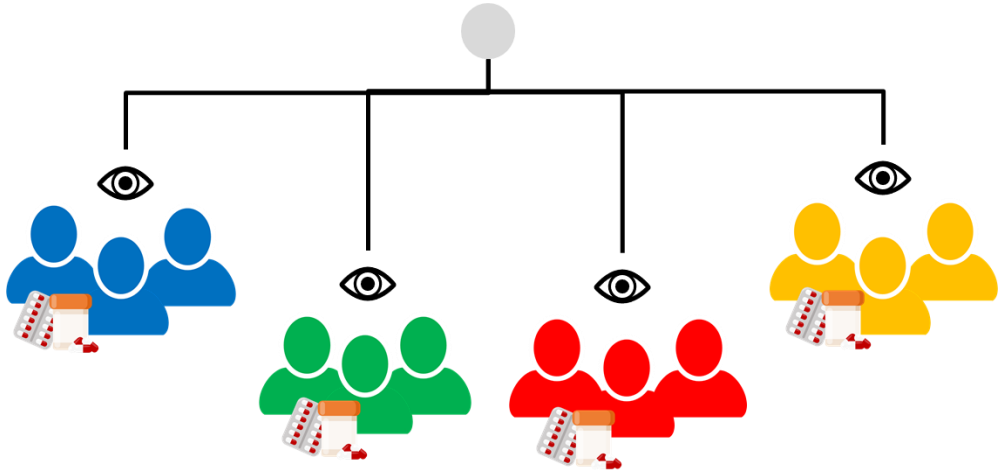
$$\mathbf{inflammation} = \mathbf{7.8} + \mathbf{1.2} \cdot \boldsymbol{smoking} + \boldsymbol{error}$$

Cross-sectional data



Cross-sectional study design

- Carried out at one time point (or short interval) to estimate the prevalence of an outcome of interest.
- Aims to describe a population, or subgroup, with respect to a set of risk factors.
- Gives no indication of sequence of events (see causal inference).
- Commonly used in for example, public health.



Levin 2006

Real-world data
non-interventional, retrospective, non-controlled, single/multi-centre study

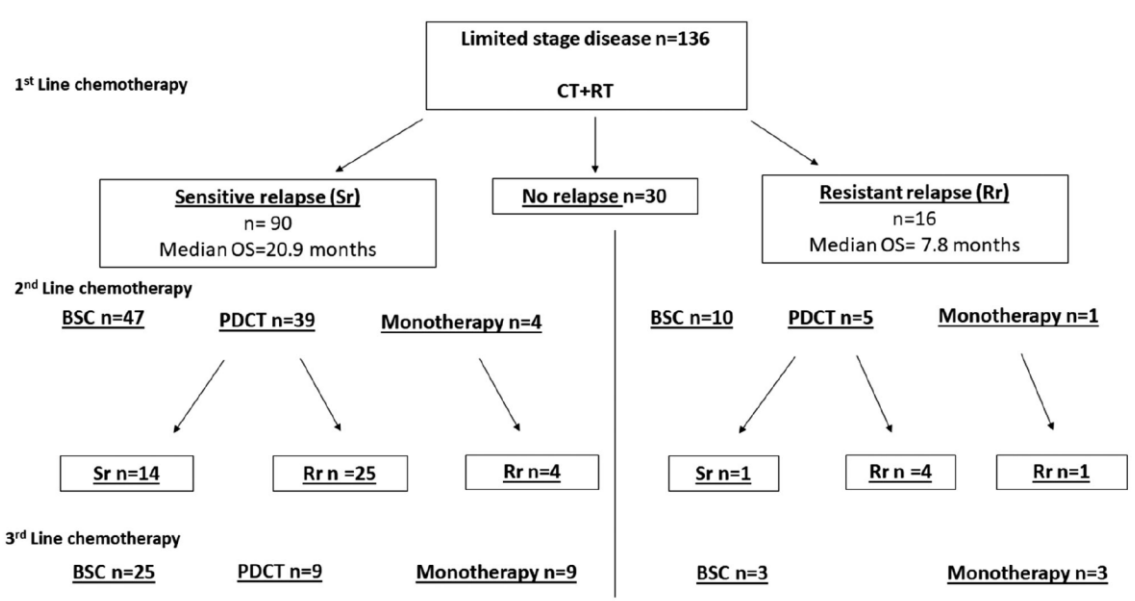


Figure 2. Treatment patterns for patients with limited disease treated with chemo- and radio-therapy (CT + RT).

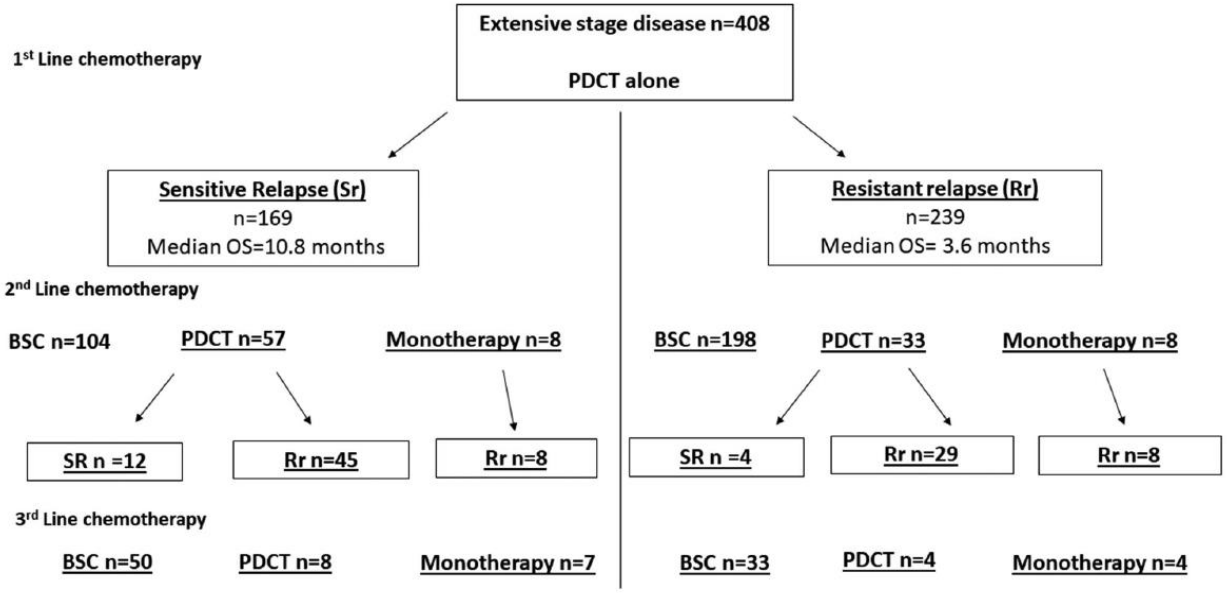


Figure 1. Treatment patterns for patients with extensive disease treated with platinum-doublet chemotherapy (PDCT alone).

Tendler *et al.* 2020

L4-3 Developing mixed-effects models



… or, approaches for how to end up with a ‘final’ model by increasing or reducing complexity.

How to carry out model development

What to consider before fitting a model:

- Study data carefully.
- **Collinearity** - combine or remove variables from the dataset?
- **Scaling of predictors and regression coefficients**, *e.g.*, height in cm vs mm vs inches vs km?
- **Standardising predictors**. For interpretability, where the intercept of *y* is the mean when *x* is at its mean.
- **Centering**. For comparison against the means of the remining independent variables.
- **Logarithmic transformation**. Additivity and linearity.
- Other transformations.

Gelman and Hill 2007

```
library(corrplot)
data.cor <- cor(data)
corrplot(data.cor)
```

How to carry out model development

Constructing the model:

- Include all input variables that, for substantive reasons, might be expected to be important in predicting the outcome.
- Inputs can be combined into total scores if needed.
- For inputs with large effect, consider also adding interaction effects.



Gelman and Hill 2007

How to carry out model development

How to remove variables:

- If a predictor is significant and has the expected sign, keep it in.
- If a predictor is not statistically significant, with the expected sign, keep in. This will not make a huge difference either way and certainly will not hurt.
- If the coefficient of a variable has the wrong expected sign, consider removing it (equivalent to setting the coefficient to zero).
- If predictor is significant with wrong sign, think hard about what this can mean.



Gelman and Hill 2007

Assessing goodness-of-fit

- **Plotting the residual errors vs. predicted values.** This gives a good indication of model bias and any heteroscedasticity (or non-uniformity of errors). It is useful to plot this for the overall model and individual groups.
- **Distribution of residuals.** Checking normality of residual errors.
- **Predicted vs. observed.** Overall and individually.
- **Validation.** Involves for example bootstrapping techniques and prediction exercises.

Model selection

- **Mechanstic reasoning.** What makes the most sense biologically/physically/ mechanistically, etc? What is the purpose,
question you are trying to answer?
- **Compare goodness-of-fit plots.** residuals vs. predictions, predicted vs. observed.
- **Likelihood ratio test.** Assess goodness-of-fit of two competing models. If there is a significant difference between two likelihoods then we reject the null model. Remember that this requires switching from REML to ML estimation procedure.
- **AIC.** The Akaike Information Criterion assesses the quality of one, or series of, models given the data set.

Likelihood ratio test

$$LR = 2\left(logL(\hat{\theta}_{ML}) - logL(\theta_{H0})\right)$$

- Maximised log-likelihood measures how well the objective function has performed, objective function value (OFV).
- A measure of relative model adequacy. The higher the better.
- If there is a significant difference between the two values (models), the ratio is relatively large and we reject the H_0 (null) model.

AIC: Akaike Information Criterion

$$AIC = N \cdot \ln\left(\frac{SS_{error}}{N}\right) + 2 \cdot K$$

- Trade-off between goodness-of-fit of the model and model complexity.

  N: number of observations,

  SS_{error}: sum of squares error,

  K: N parameters + 1

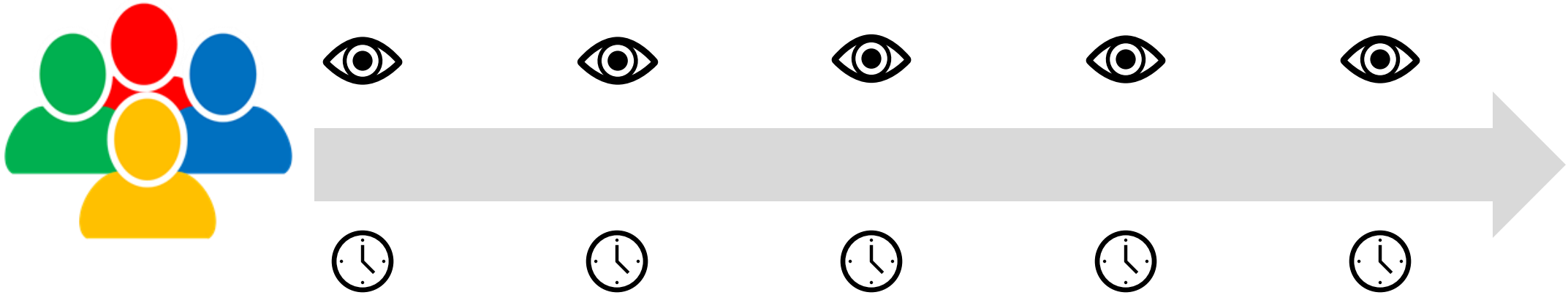
- The lower the value of AIC the better is the goodness-of-fit relative to model complexity.
- Bayesian Information Criterion (BIC): penalises N parameters to a higher extent than the AIC.

Akaike 1973



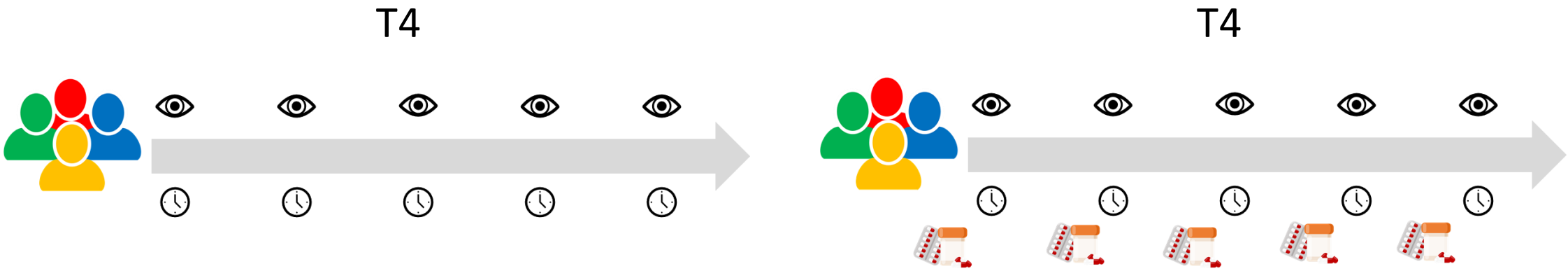
figure: https://www.verywellhealth.com/overview-of-hypothyroidism-4164534

Study design



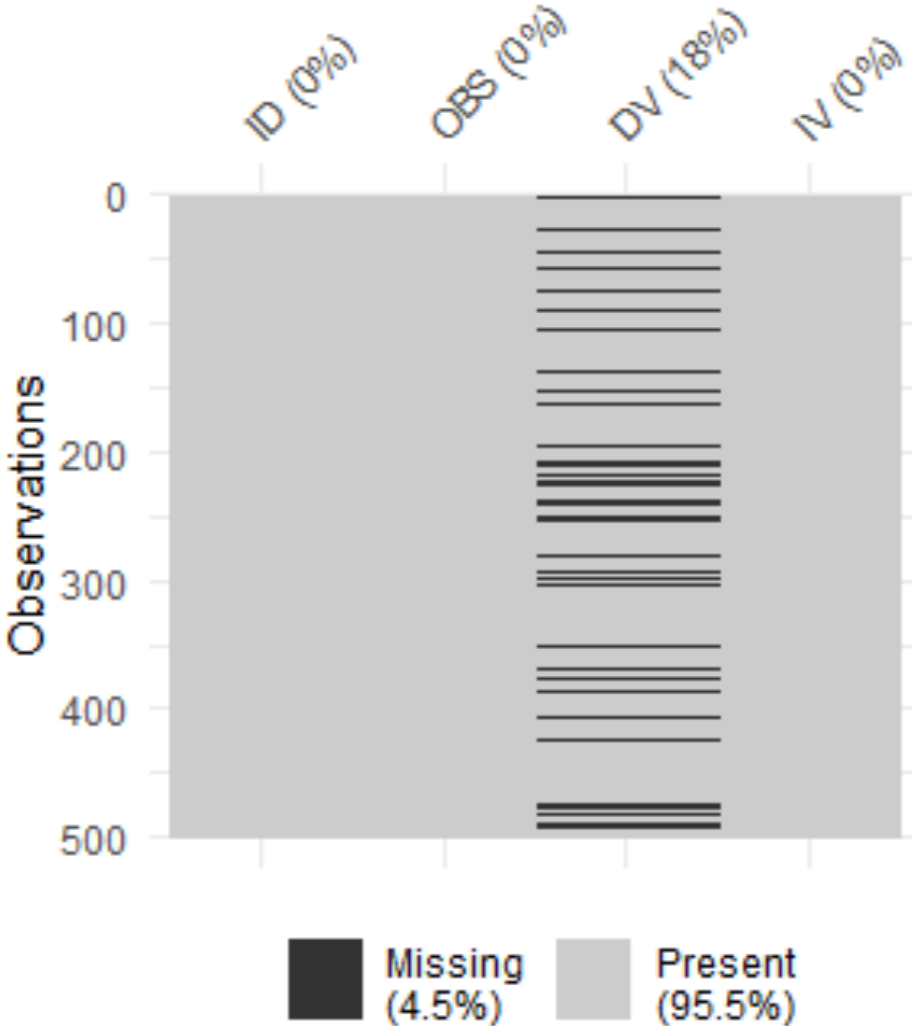
Study design and dataset

ID: subject ID,
OBS: time point of observation,
DV: dependent variable, treatment effect,
IV: intervention - drug treatment,
~5% random missingness

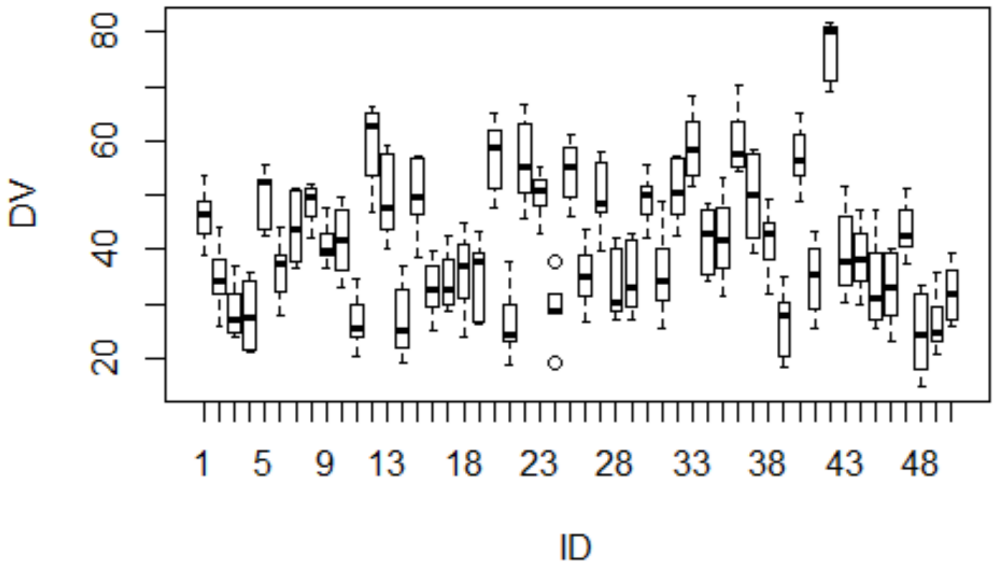


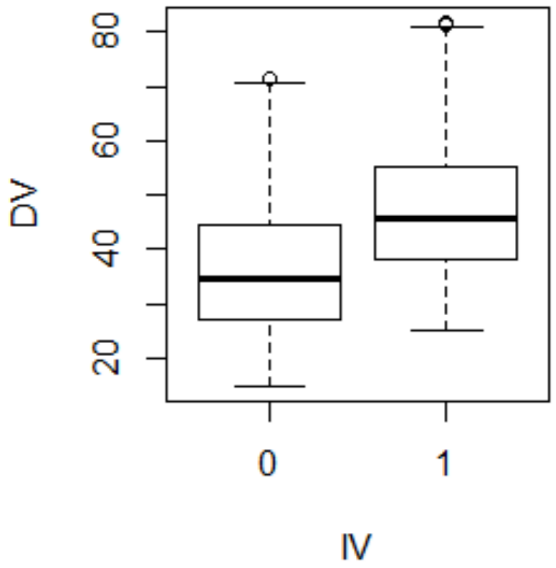
treatment dosed to steady state

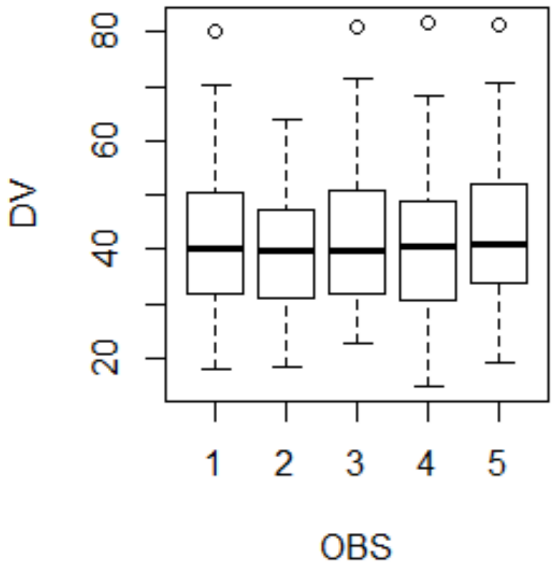
Missingness



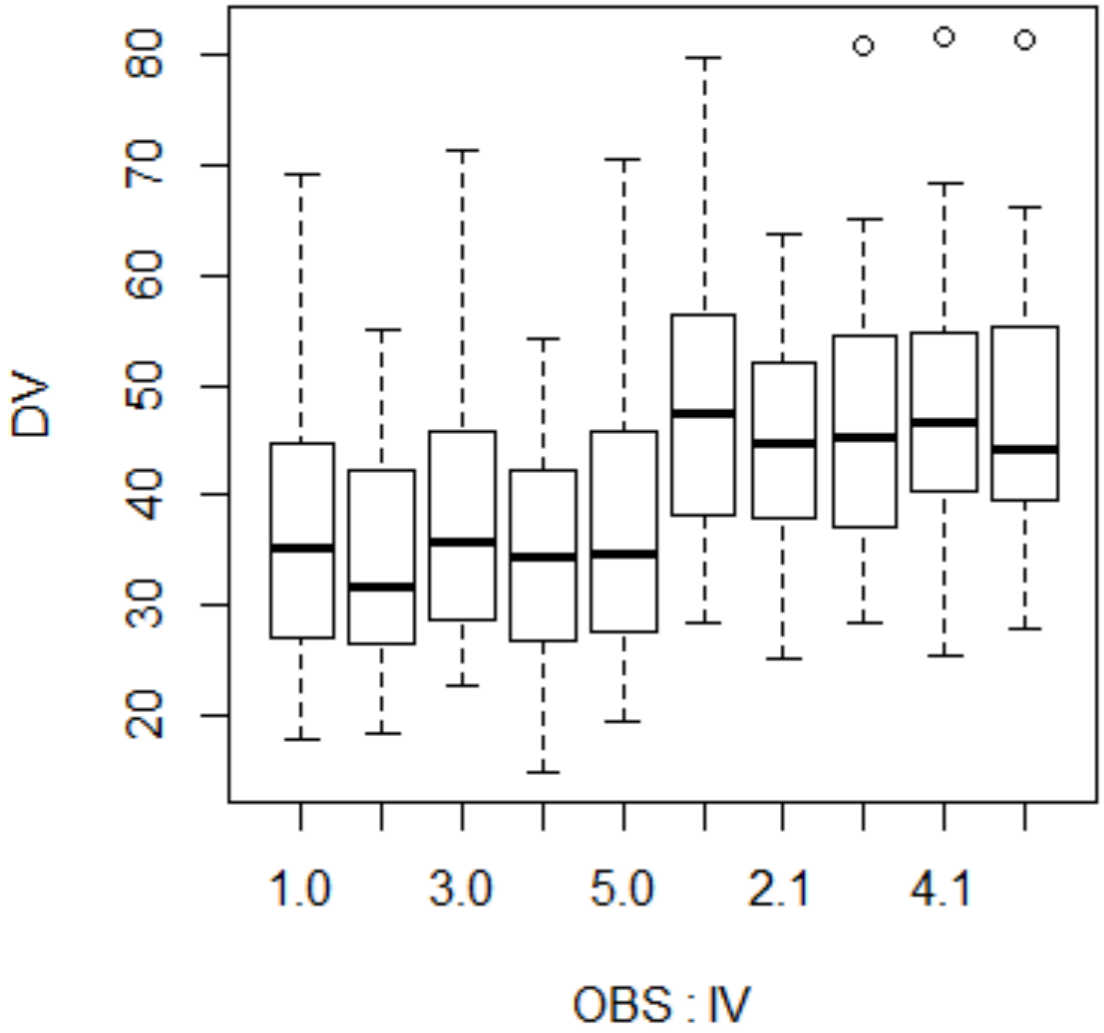

```
library(naniar)
vis_mis(exdata)
```


DV
20
40
60
80
1
5
9
13
18
23
28
33
38
43
48
ID


DV
20
40
60
80
0
1
IV

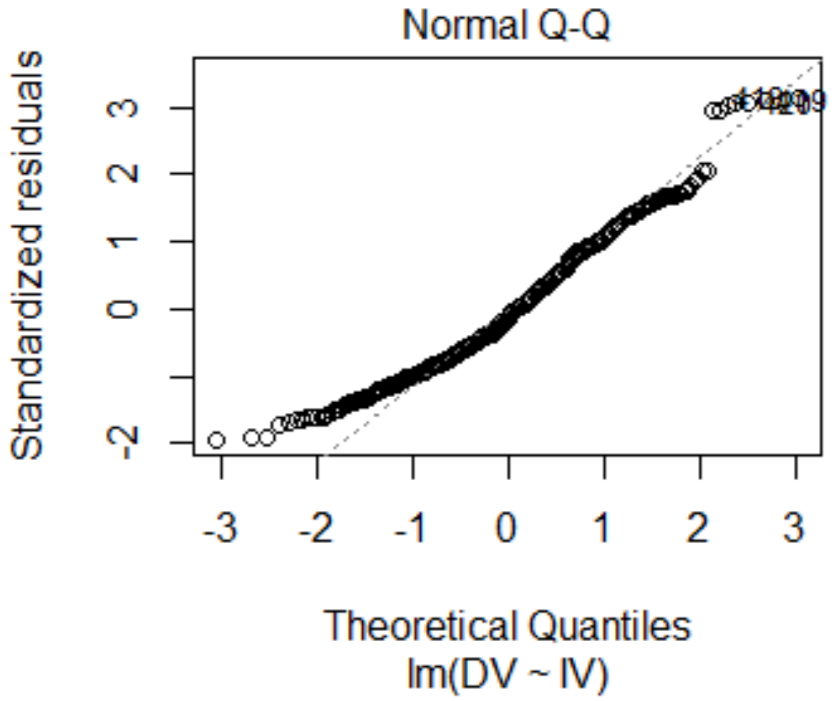

DV
20
40
60
80
1
2
3
4
5
OBS

```
# boxplots
boxplot(DV ~ ID, data=exdata)
boxplot(DV ~ ID + IV, data=exdata)
boxplot(DV ~ OBS, data=exdata)
boxplot(DV ~ OBS + IV, data=exdata)
```



fixed-effects model: complete pooling

```
# fixed effects fit: complete
# pooling
m1_fe1 <- lm(formula = DV ~ IV)
plot(m1_fe1)
```



```
> summary(m1_fe1)

Call:
lm(formula = DV ~ IV)

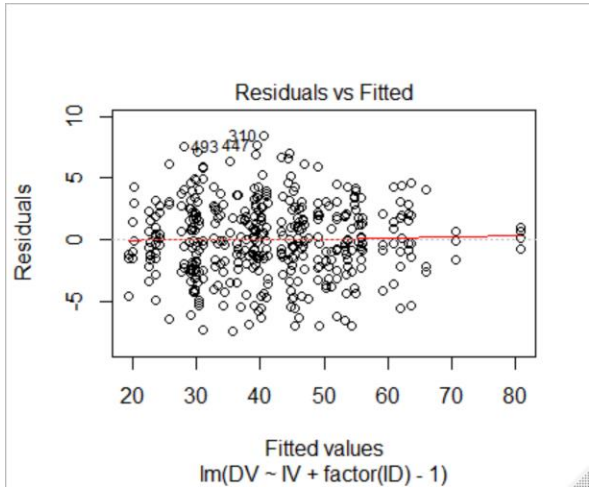
Residuals:
    Min      1Q  Median      3Q     Max
-21.796  -8.863  -1.488   8.399  35.142

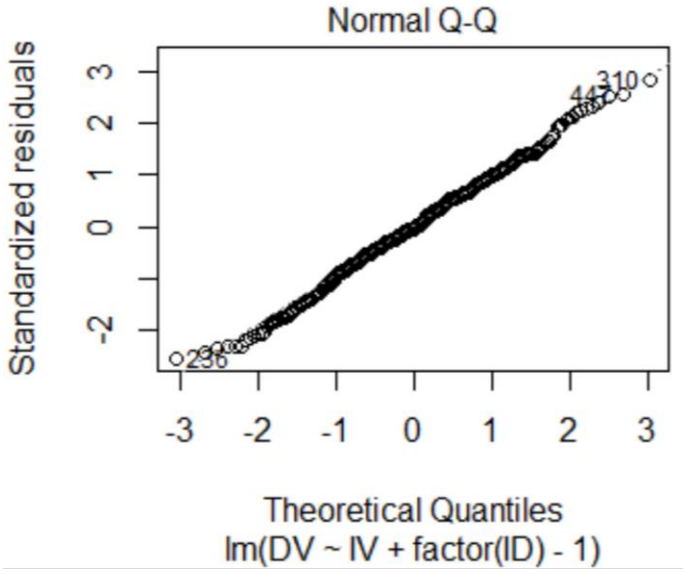
Coefficients:
            Estimate Std. Error t value
(Intercept)  36.1619     0.7814  46.277
IV           10.7020     1.1133   9.613
            Pr(>|t|)
(Intercept)   <2e-16 ***
IV            <2e-16 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05
  '.' 0.1 ' ' 1

Residual standard error: 11.27 on 408 degrees of freedom
  (90 observations deleted due to missingness)
Multiple R-squared:  0.1847,     Adjusted R-squared:  0.1827
F-statistic: 92.41 on 1 and 408 DF,  p-value: < 2.2e-16
```

fixed-effects model: no pooling

```
# fixed effects fit: no
# pooling
m2_fe2 <- lm(formula = DV ~ IV +
factor(ID) - 1)
plot(m2_fe2)
```





```
> summary(m2_fe2)

Call:
lm(formula = DV ~ IV + factor(ID) - 1)

Residuals:
    Min      1Q  Median      3Q     Max
-7.4353 -1.9040 -0.1054  1.9683  8.4838

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
IV            10.0337     0.3199   31.36   <2e-16 ***
factor(ID)1   40.3414     1.0792   37.38   <2e-16 ***
factor(ID)2   29.7712     1.0740   27.72   <2e-16 ***
factor(ID)3   24.2284     1.0740   22.56   <2e-16 ***
...
factor(ID)50  28.1311     1.1354   24.77   <2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.194 on 359 degrees of freedom
  (90 observations deleted due to missingness)
Multiple R-squared:  0.9952,     Adjusted R-squared:  0.9946
F-statistic:  1468 on 51 and 359 DF,  p-value: < 2.2e-16
```

Mixed-effects model: null

```
# mixed-effects model null
# pooling
m0_lme <- lmer(DV ~ 1 + (1 | ID), exdata)
summary(m0_lme)
summ(m0_lme)
ranova(m0_lme)
plot(m0_lme)
```

Mixed-effects model: treatment

```
# mixed-effects model treatment
# pooling
m1_lme <- lmer(DV ~ 1 + IV + (1 | ID), exdata)
summary(m1_lme)
summ(m1_lme)
ranova(m1_lme)
plot(m1_lme)
```

R

```
> summary(m1_lme)
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: DV ~ 1 + IV + (1 | ID)
   Data: exdata

REML criterion at convergence: 2340.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.3760 -0.6038 -0.0386  0.6275  2.6359

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 122.4    11.062
 Residual              10.2     3.194
Number of obs: 410, groups:  ID, 50

Fixed effects:
            Estimate Std. Error       df t value Pr(>|t|)
(Intercept)  36.4495     1.5803  49.9485   23.07   <2e-16 ***
IV           10.0421     0.3199 359.1928   31.39   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
   (Intr)
IV -0.100
```

R

```
> summ(m1_lme)
MODEL INFO:
Observations: 410
Dependent Variable: DV
Type: Mixed effects linear regression

MODEL FIT:
AIC = 2348.93, BIC = 2364.99
Pseudo-R² (fixed effects) = 0.16
Pseudo-R² (total) = 0.94

FIXED EFFECTS:
----------------------------------------------------------
                      Est.   S.E.   t val.      d.f.      p
----------------- ------- ------ -------- -------- ------
(Intercept)          36.45   1.58    23.07    49.95   0.00
IV                   10.04   0.32    31.39   359.19   0.00
----------------------------------------------------------

p values calculated using Satterthwaite d.f.

RANDOM EFFECTS:
-----------------------------------
  Group      Parameter    Std. Dev.
---------- ------------- -----------
    ID       (Intercept)     11.06
 Residual                    3.19
-----------------------------------

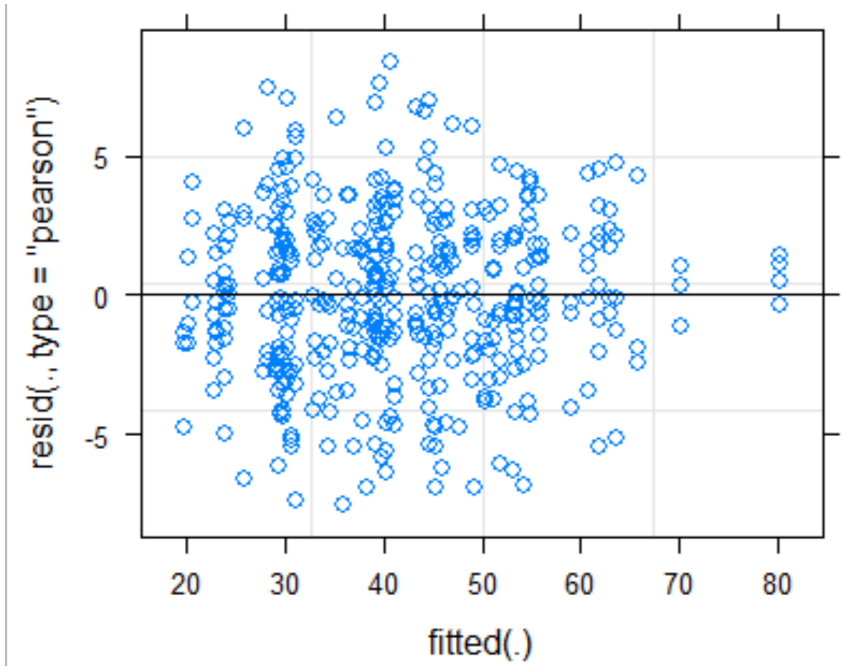
Grouping variables:
-------------------------
 Group   # groups   ICC
------- ---------- ------
  ID        50      0.92
-------------------------
```

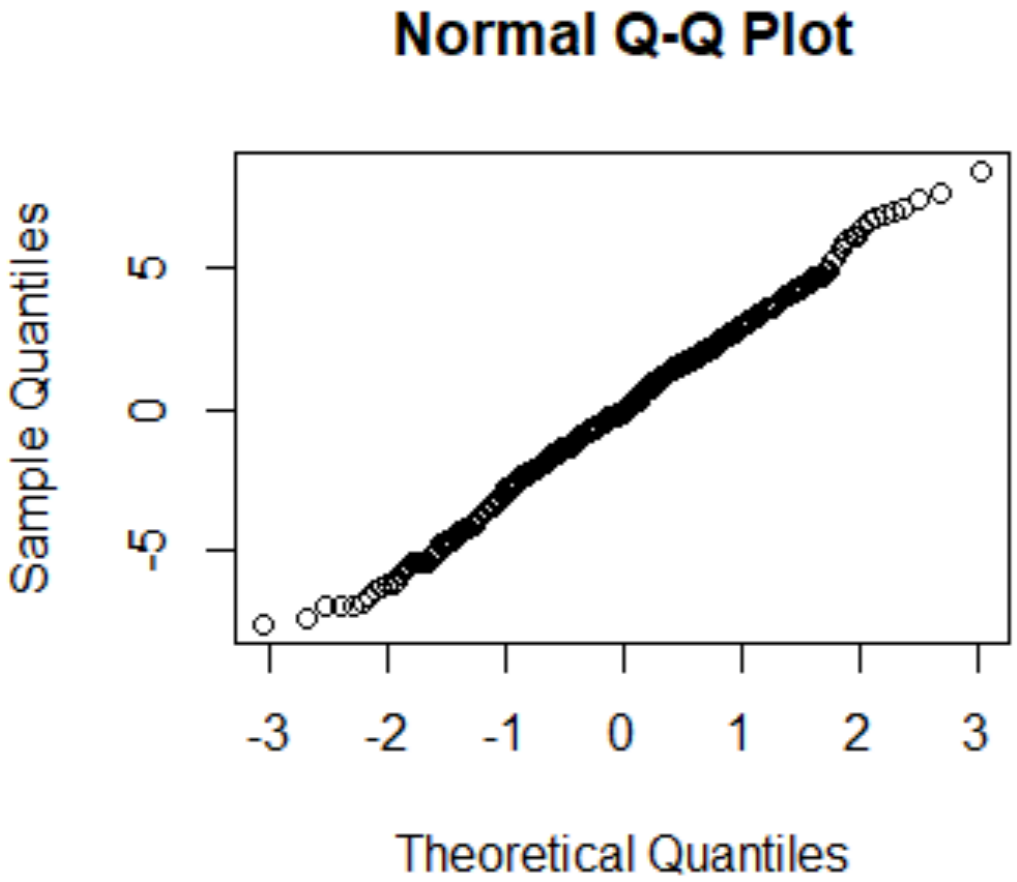
mixed-effects model: likelihood ratio test

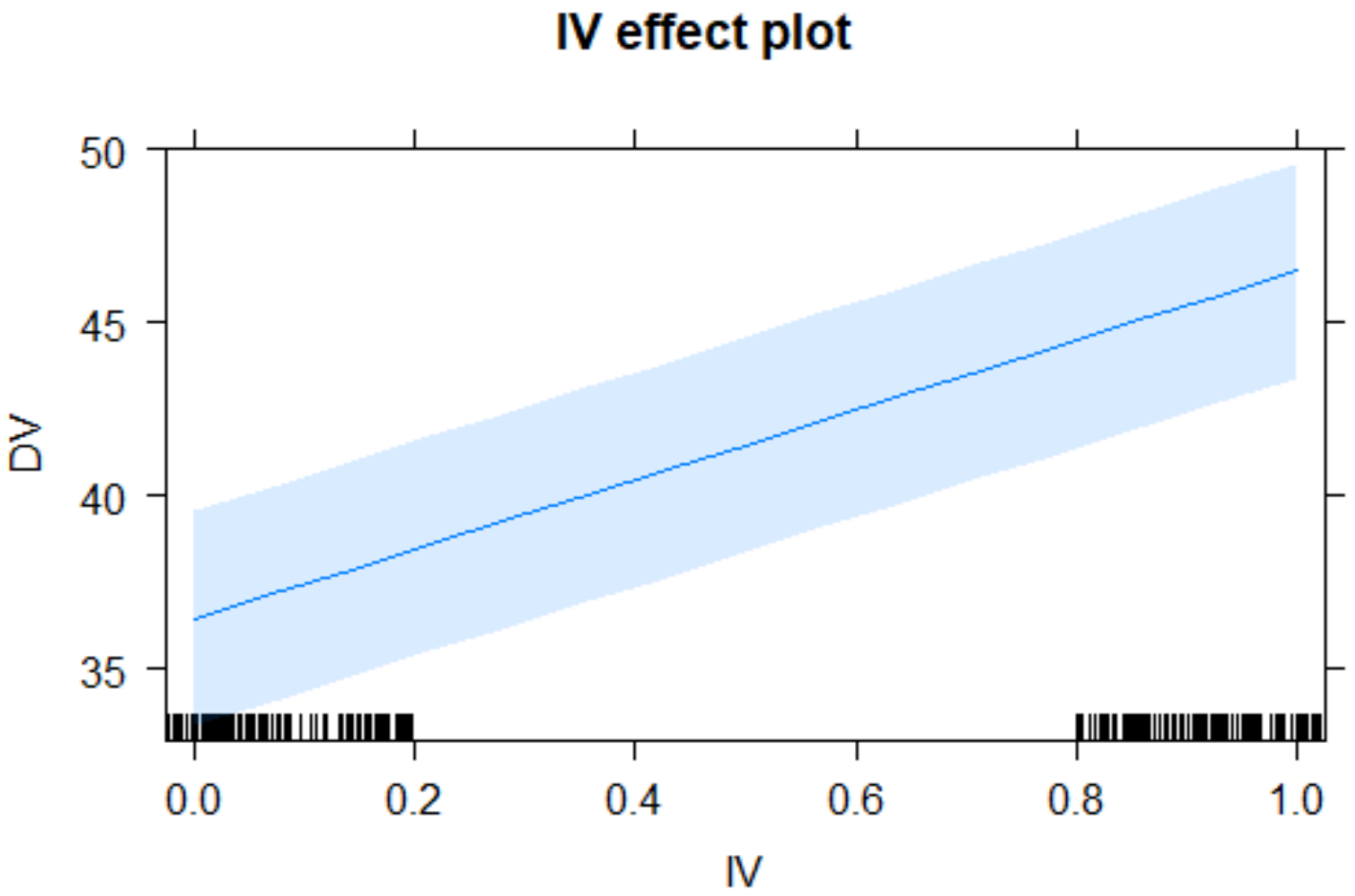
```
# likelihood ratio test
m_null <- lmer(DV ~ 1 + (1 | ID), exdata, REML = FALSE)
m_treat <- lmer(DV ~ 1 + IV + (1 | ID), exdata, REML = FALSE)
anova(m_null,m_treat)
```

R

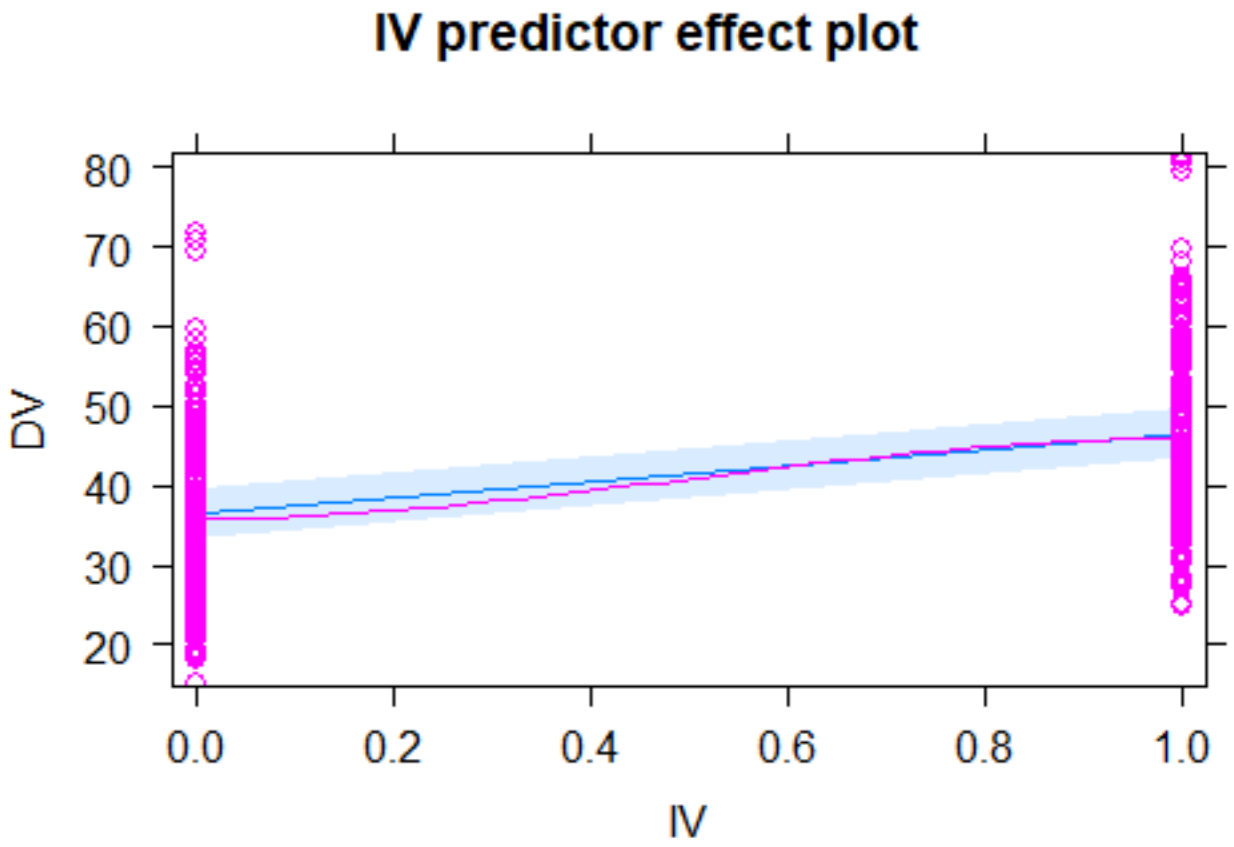
```
> anova(m_null,m_treat)
Data: exdata
Models:
m_null: DV ~ 1 + (1 | ID)
m_treat: DV ~ 1 + IV + (1 | ID)
        npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
m_null     3 2826.0 2838.0 -1410.0   2820.0
m_treat    4 2351.2 2367.3 -1171.6   2343.2 476.74  1  < 2.2e-16
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```


resid(., type = "pearson")
5
0
-5
20
30
40
50
60
70
80
fitted(.)


Normal Q-Q Plot
Sample Quantiles
5
0
-5
-3
-2
-1
0
1
2
3
Theoretical Quantiles




```
# plot effects
Plot(allEffects(m1_lme))
```




```
plot(predictorEffects(m1_lme,~IV,
partial.residuals=TRUE))
```

Bootstrap

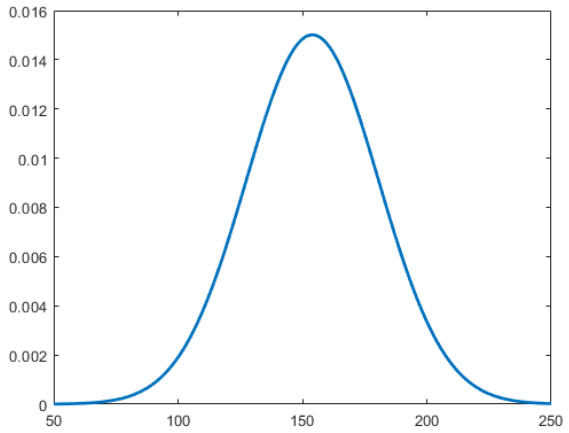
```
> library(boot)
> confint(m2_lme,parm=c(3,4),method="boot",nsim=1000,boot.type="perc")
Computing bootstrap confidence intervals ...
                2.5 %   97.5 %
(Intercept) 33.28305 39.58972
IV           9.39840 10.62247
```

Summary

- We reviewed:
  - The limits of fixed effects linear regression modelling.
  - When and why you should consider mixed-effects/multi-level modelling.
  - The theory behind mixed-effects modelling.
  - A basic workflow for mixed-effects modelling to get you started, learn more and perhaps inform your own opinions (few right and wrong answers).
  - A couple of practical examples and case studies.
- By properly accounting for variability, we are better able to estimate effects.
- Mixed-effects modelling can do far more than what we covered today.
- This is a powerful approach for unravelling effects hidden away behind variability.

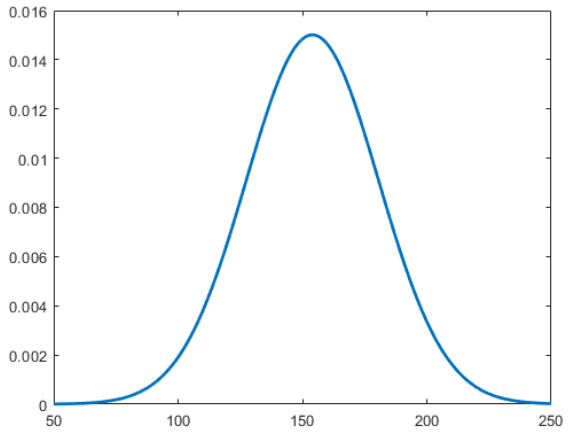
L4-4 Maximum Likelihood (ML) Estimation & REML

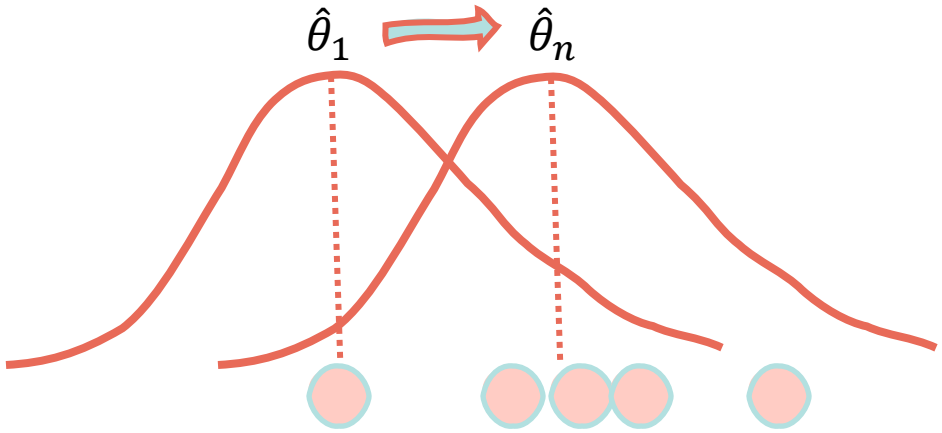
Maximum likelihood (ML) estimation



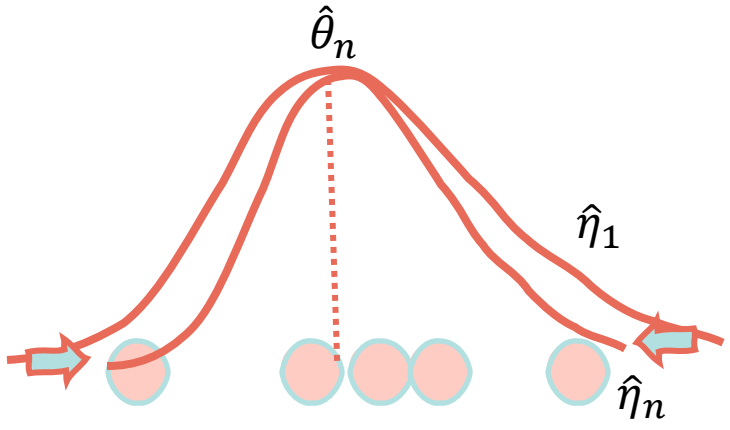
- Central to much of model estimation-based statistics.
- Estimate the mean then estimate the variance of the mean.
- Given a dataset $M = [X, Y]$ and a model parameter θ.
  What is the probability of observing the data given the parameter likelihood? $L(\theta|D)$
- Maximise the log likelihood: $\hat{\theta} = argmax\ L(\theta|M)$
- Or minimise the negative log likelihood ($-LL$).

Maximum likelihood (ML) estimation





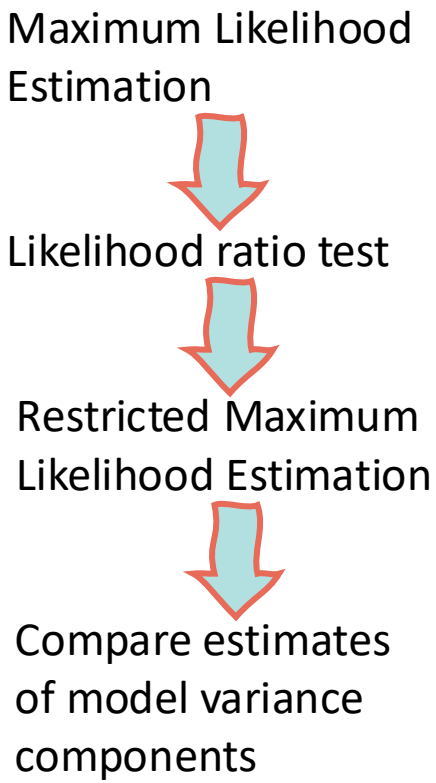
- Fitting a normal distribution for a given parameter, *e.g.* β_i.
- I: What is the mean that maximises the likelihood of observing the dependent variable?
- II: Given the estimated mean, what is the standard deviation that maximises the likelihood of observing the dependent variable?
- θ, ML estimate of the mean. η, ML estimate of the stdev.
- More data tend to mean a more accurate estimate.



Restricted/residual maximum likelihood estimation (REML)

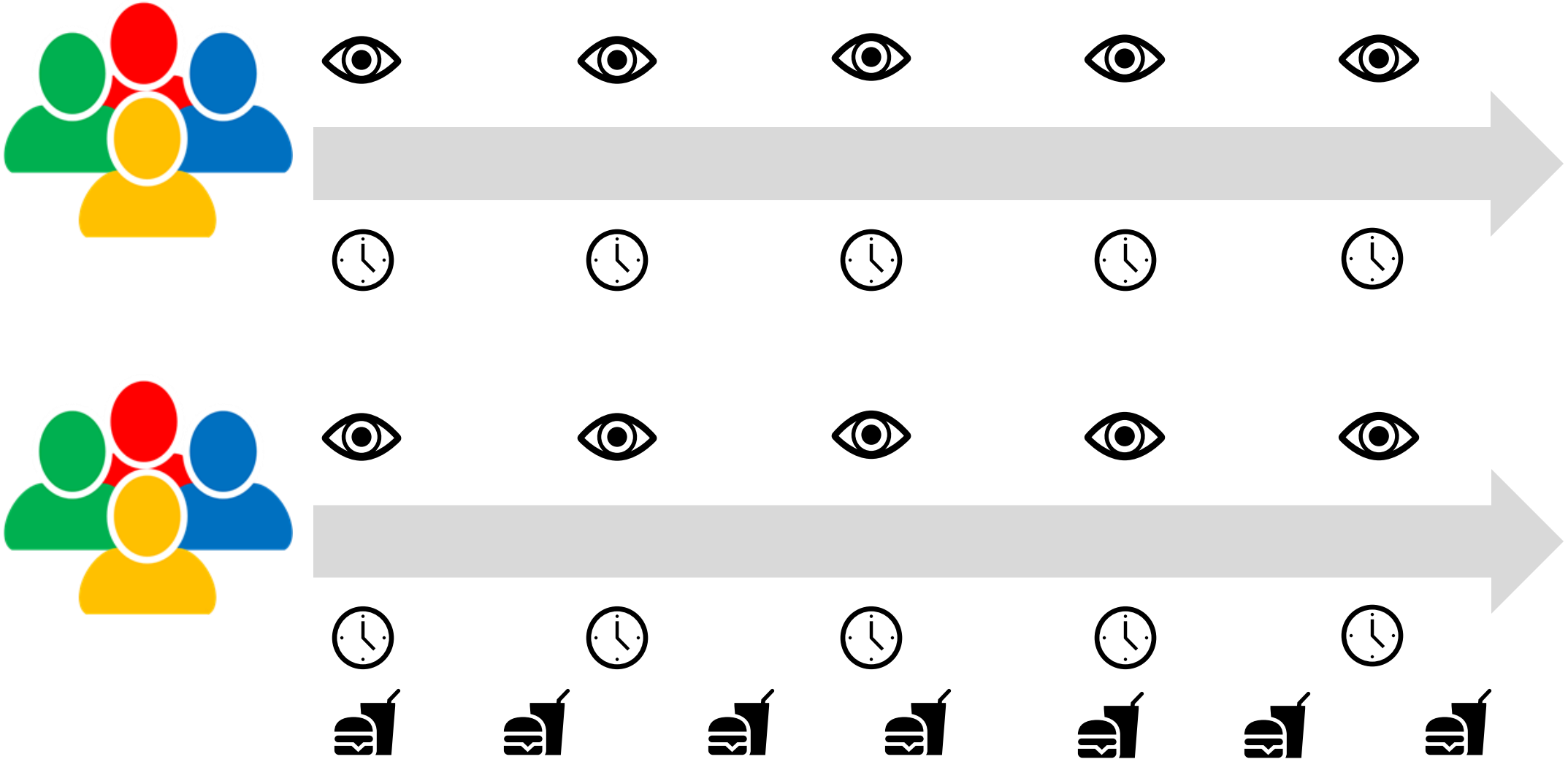
```
m_null <- lmer(DV ~ 1 + (1 | ID), exdata,
REML = FALSE/TRUE)
```

- In small sample sizes the ML produces biased estimates of the standard deviation of the variance (usually inflated).
- REML is a variation of normal likelihood.
- Corrects (to some extent) for the bias in the variance component.
- Less biased in small samples relative to ML.
- Cannot use in likelihood ratio test to compare models.

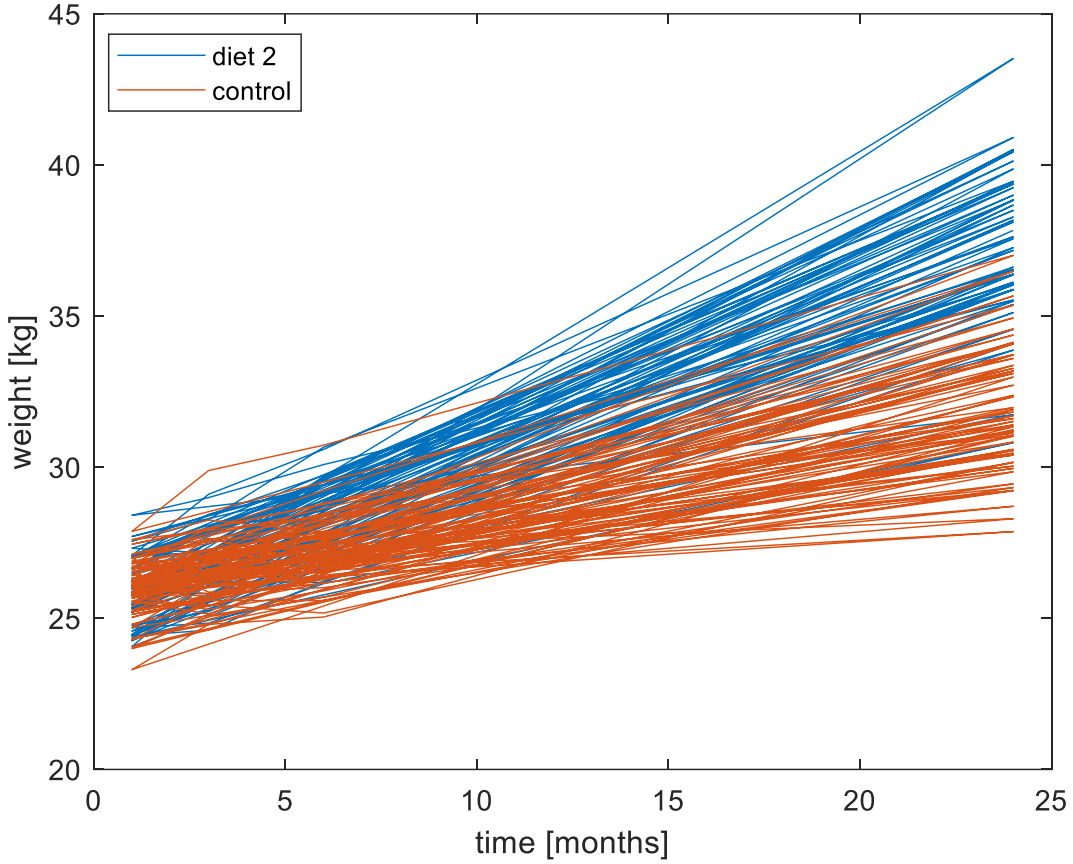


L4-5 Developing Linear Mixed-Effects Regression Models

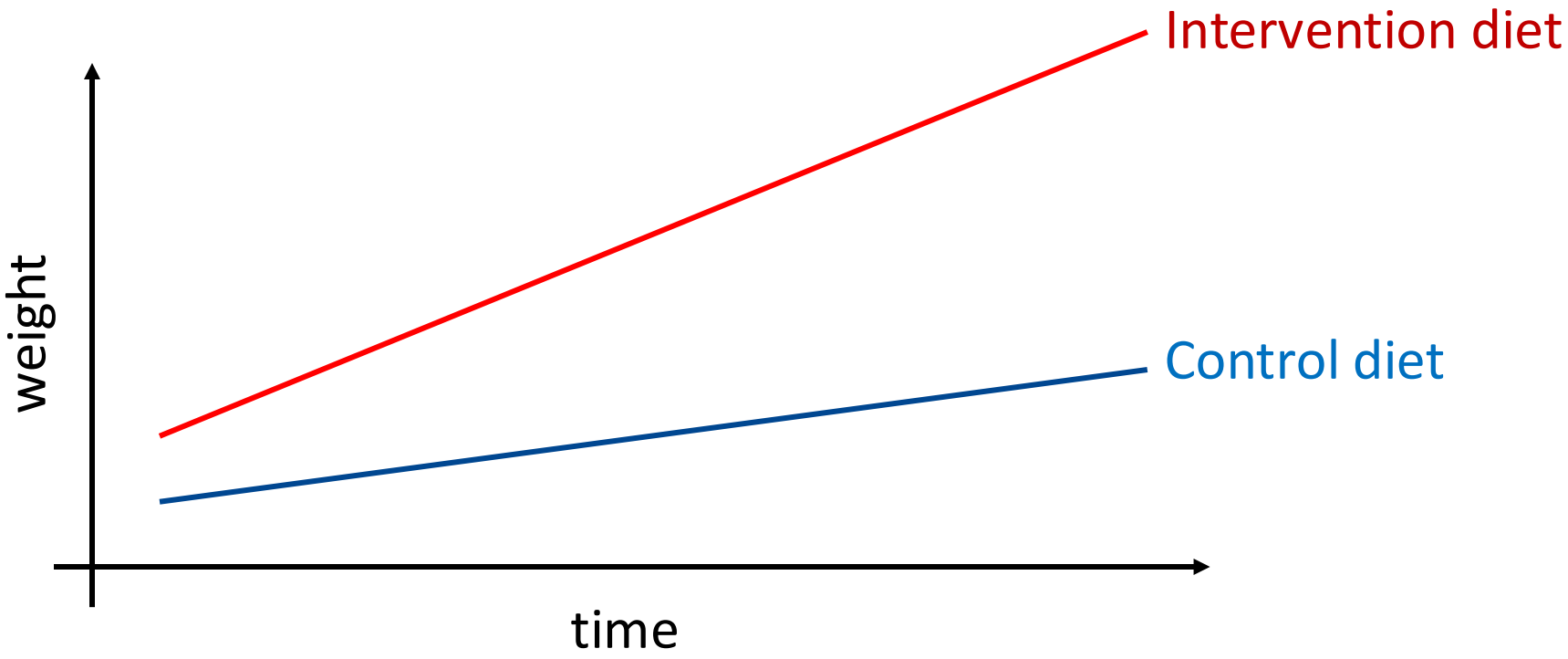
The Paediatric Study diet and body weight



The Paediatric Study diet and body weight



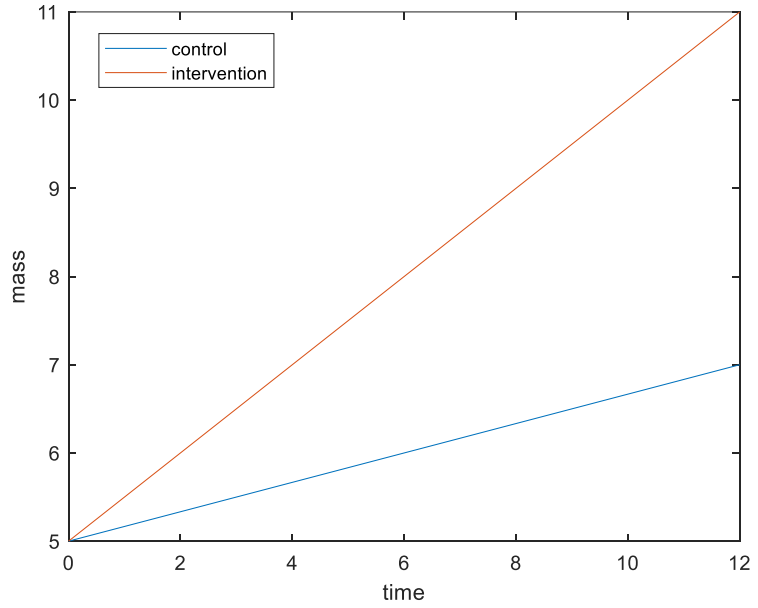
Interaction terms



$$Y = \beta_0 + \beta_1 INT + \beta_2 time + \varepsilon \qquad Y = \beta_0 + \beta_1 INT + \beta_2 time + \beta_3\ INT\ time + \varepsilon$$

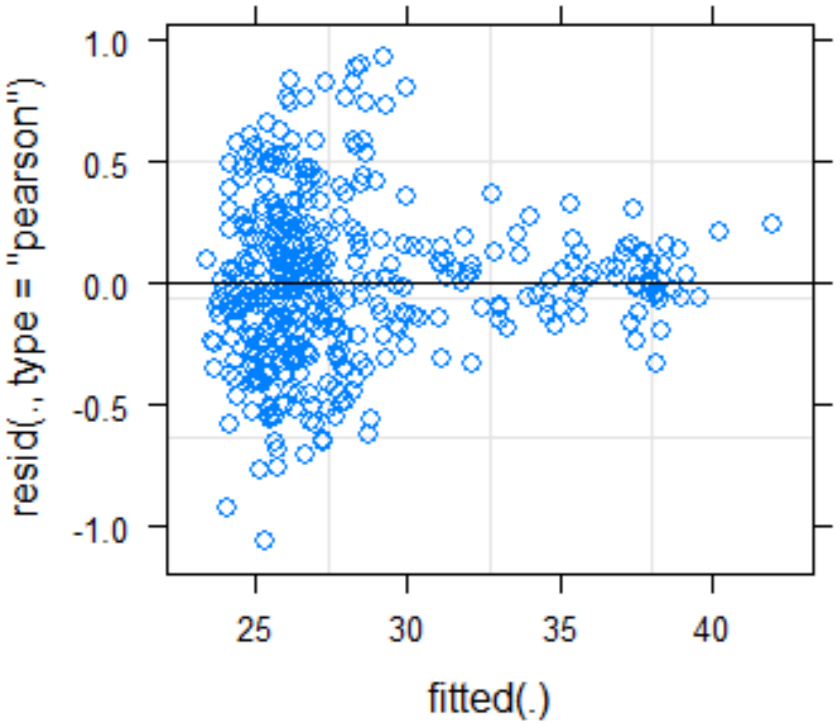
- The effect of an independent variable on the dependent variable is dependent on another independent variable.
- When investigate: based on mechanism, or effect size.

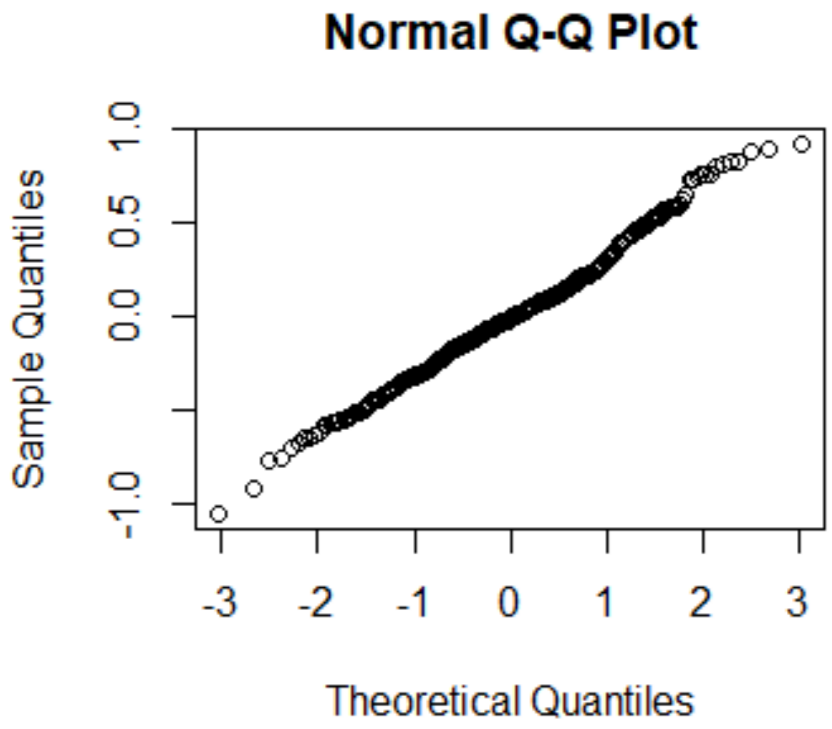
Conditional growth model

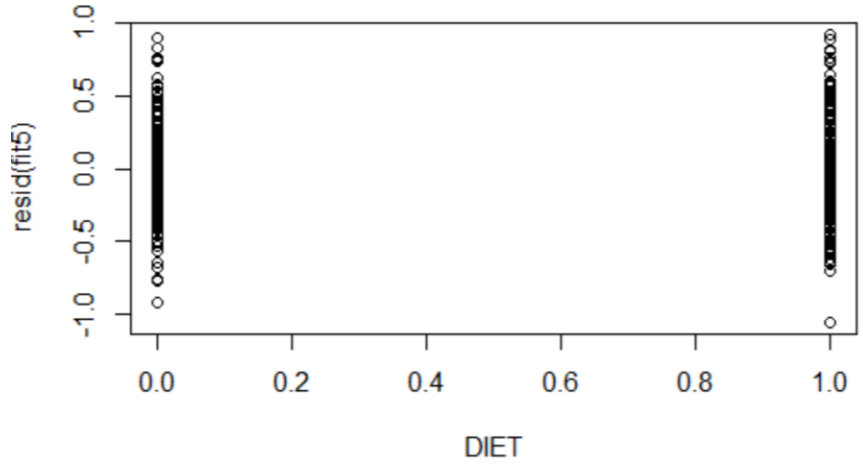


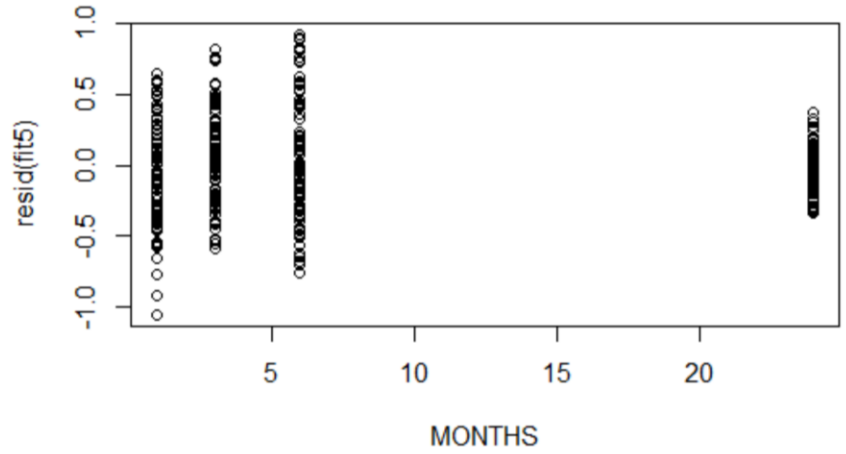
$$Y = \beta_0 + \beta_1 \cdot time + \varepsilon$$

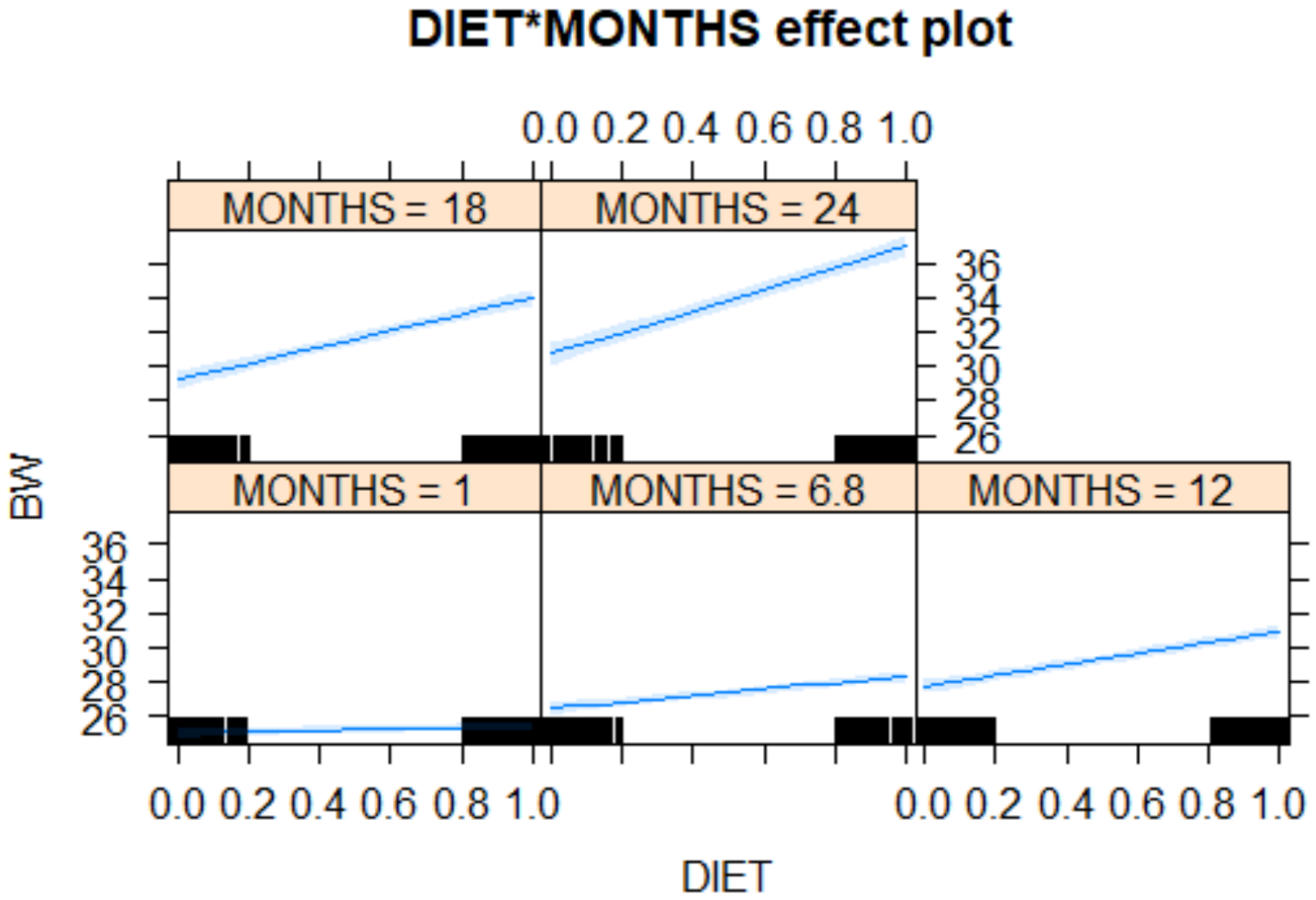
```
# conditional growth model
model <- lmer(BW ~ 1 + Months*Diet + (1 + Months | ID),
data)
```


resid(., type = "pearson")
1.0
0.5
0.0
-0.5
-1.0
25
30
35
40
fitted(.)


Normal Q-Q Plot
Sample Quantiles
1.0
0.5
0.0
-1.0
-3
-2
-1
0
1
2
3
Theoretical Quantiles


resid(fit5)
1.0
0.5
0.0
-0.5
-1.0
0.0
0.2
0.4
0.6
0.8
1.0
DIET


resid(fit5)
1.0
0.5
0.0
-0.5
-1.0
5
10
15
20
MONTHS




```
# plot effects
Plot(allEffects(model))
```

```
> summary(model)
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: BW ~ DIET * MONTHS + (MONTHS | ID)
   Data: dietdata

REML criterion at convergence: 1032.4

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.37610 -0.47912 -0.01658  0.41414  2.08534

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 ID       (Intercept) 0.877062 0.93652
          MONTHS      0.005634 0.07506  0.05
 Residual             0.196379 0.44315
Number of obs: 400, groups:  ID, 100

Fixed effects:
            Estimate Std. Error       df t value Pr(>|t|)
(Intercept) 24.80192    0.14838 98.01430 167.149   <2e-16 ***
DIET         0.14432    0.19828 98.01430   0.728    0.468
MONTHS       0.24656    0.01189 97.99779  20.731   <2e-16 ***
DIET:MONTHS  0.25396    0.01589 97.99779  15.980   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) DIET   MONTHS
DIET        -0.748
MONTHS      -0.015  0.011
DIET:MONTHS  0.011 -0.015 -0.748
```

```
> summ(model)
MODEL INFO:
Observations: 400
Dependent Variable: BW
Type: Mixed effects linear regression

MODEL FIT:
AIC = 1048.41, BIC = 1080.34
Pseudo-R² (fixed effects) = 0.88
Pseudo-R² (total) = 0.99

FIXED EFFECTS:
----------------------------------------------------------
                     Est.   S.E.   t val.     d.f.      p
----------------- ------- ------ -------- ------- ------
(Intercept)         24.80   0.15   167.15   98.01   0.00
DIET                 0.14   0.20     0.73   98.01   0.47
MONTHS               0.25   0.01    20.73   98.00   0.00
DIET:MONTHS          0.25   0.02    15.98   98.00   0.00
----------------------------------------------------------

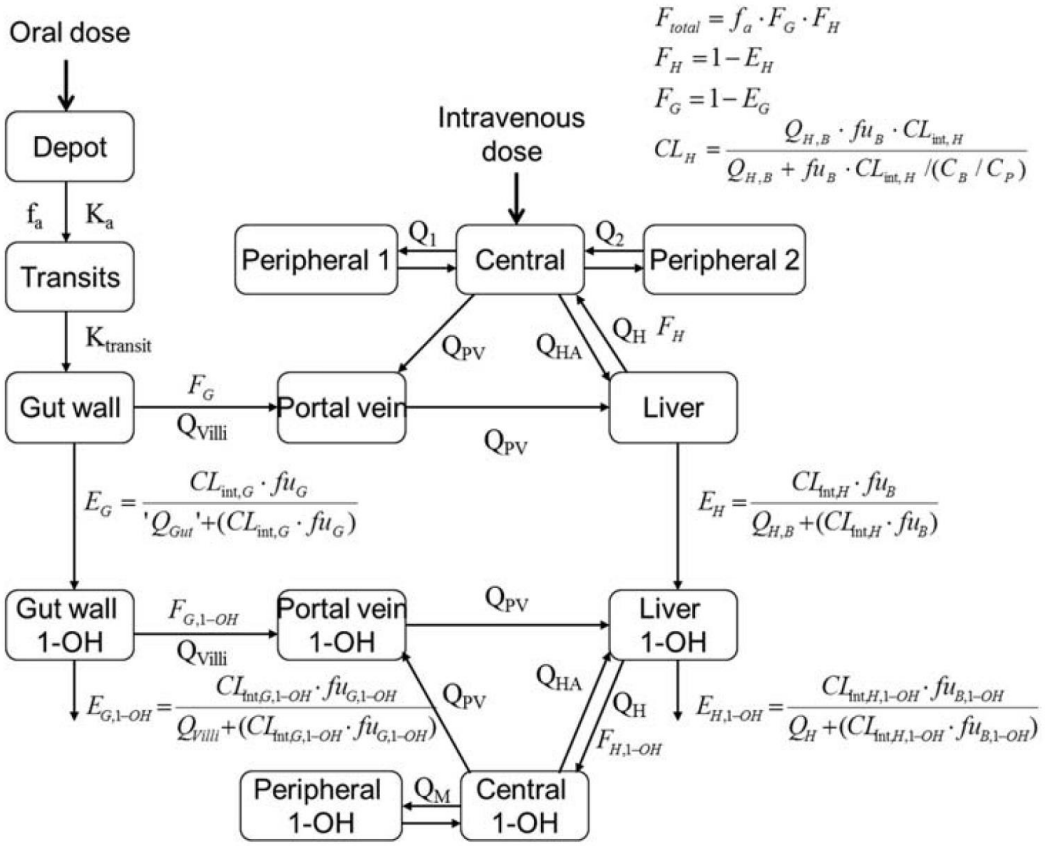
p values calculated using Satterthwaite d.f.

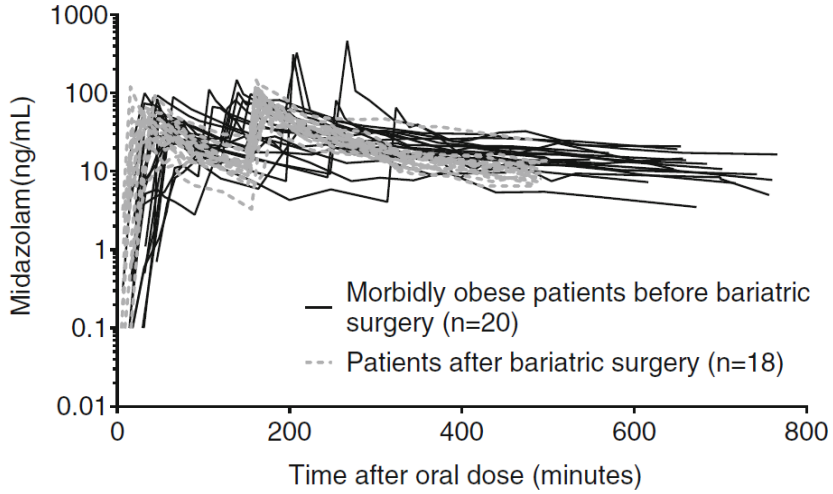
RANDOM EFFECTS:
-----------------------------------
  Group      Parameter    Std. Dev.
--------- ------------- -----------
    ID      (Intercept)     0.94
    ID        MONTHS        0.08
 Residual                   0.44
-----------------------------------

Grouping variables:
-------------------------
 Group   # groups   ICC
------- ---------- ------
  ID       100      0.82
-------------------------
```

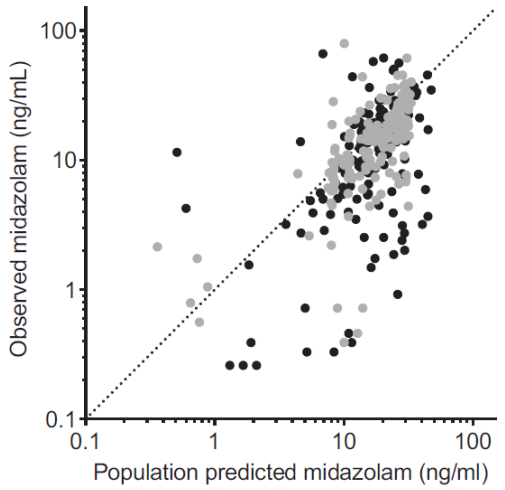
Semiphysiologically Based Pharmacokinetic Model for Midazolam and CYP3A Mediated Metabolite 1-OH-Midazolam in Morbidly Obese and Weight Loss Surgery Patients

MJE Brill[1,2], PAJ Välitalo[1], AS Darwich[3], B van Ramshorst[4], HPA van Dongen[5], A Rostami–Hodjegan[3], M Danhof[1] and CAJ Knibbe[1,2]*

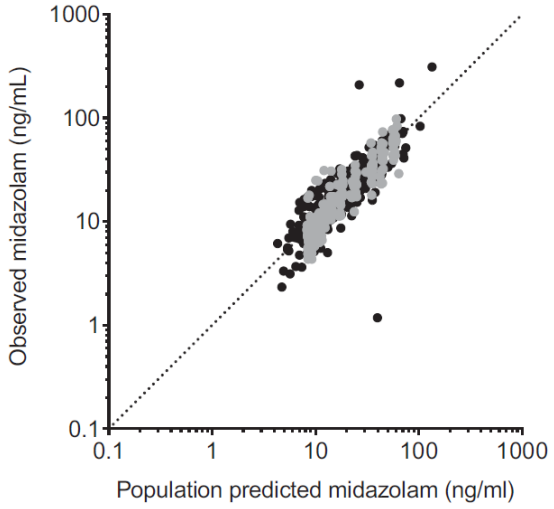




After oral dose



After intravenous dose

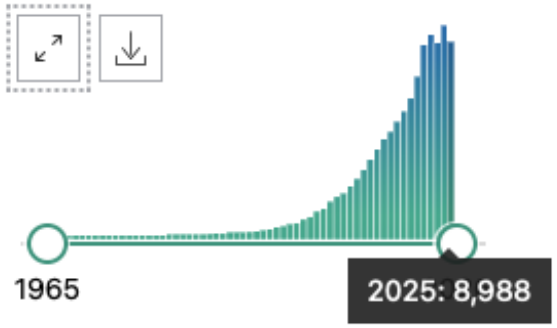


L4-6 Mixed-effects modelling and study design (II) optional viewing

Nonlinear mixed-effects models
clinical study design and modelling

PubMed®

RESULTS BY YEAR



FDA U.S. FOOD & DRUG ADMINISTRATION

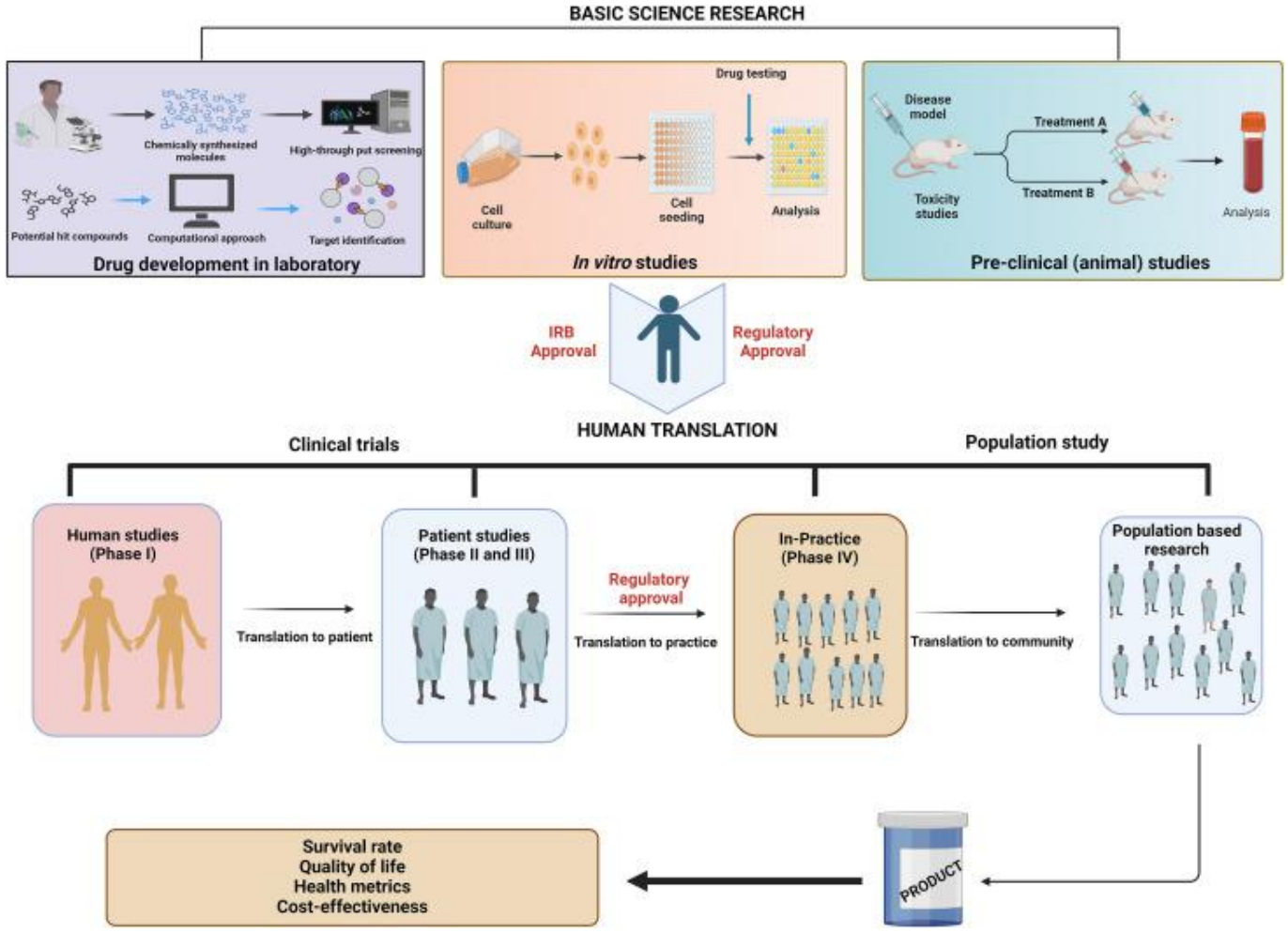
B. The Nonlinear Mixed-Effects Modeling Approach

When properly performed, population PK studies in patients combined with suitable mathematical/statistical analysis, for example, using nonlinear mixed-effects modeling, is a valid, and on some occasions, preferred alternative to extensive studies. In sparse data situations, where the traditional two-stage approach is not applicable because estimates of individual parameters are, a priori, out of reach, a single-stage approach, such as nonlinear mixed-effects modeling, should be used.

In the context of drug evaluation, the nonlinear mixed-effects modeling approach developed from the recognition that, if pharmacokinetics and pharmacodynamics are to be investigated in patients, pragmatic considerations dictate that data should be collected under less stringent and restrictive design conditions. This approach considers the

[4] Other approaches, such as the naive averaged-data approach, which provides mean population pharmacokinetic parameter estimates without variability estimates, are available for use, but will not be discussed.

Nonlinear mixed-effects models
clinical study design and modelling



El-Tanani *et al*. 2025

Nonlinear mixed-effects models
clinical study design and modelling

DOI: 10.1111/pan.13548

RESEARCH REPORT

WILEY Pediatric Anesthesia

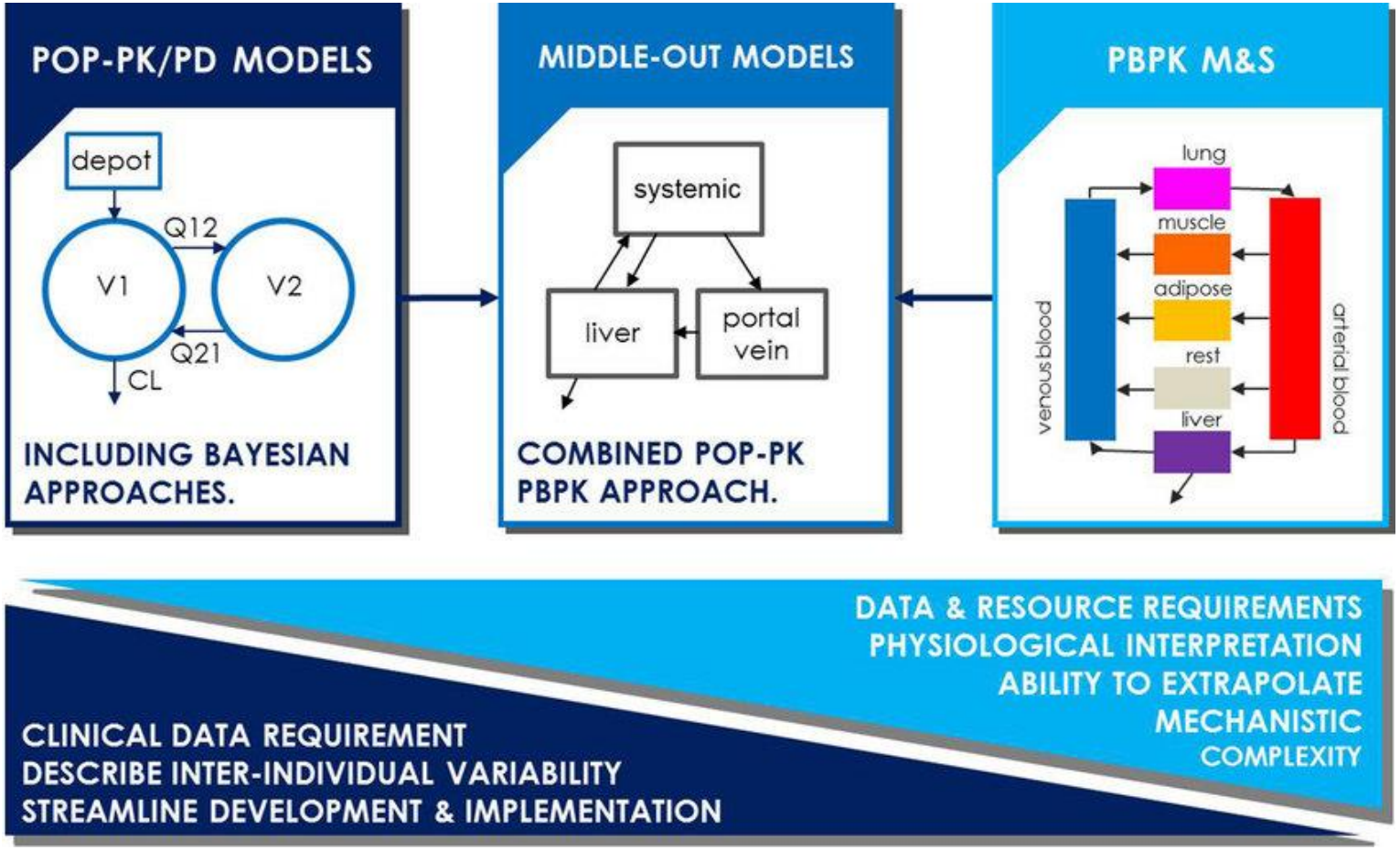
A study of the dosage and duration for levobupivacaine infusion by the caudal-epidural route in infants aged 3-6 months

Rita Vashisht[1] | Anju A. Bendon[1] | Ijeoma Okonkwo[1] | Davandra Patel[1] | Catherine Fullwood[2] | Kayode Ogungbenro[3] | Leon Aarons[3] | Adam S. Darwich[3]

- Only short duration data in paediatrics.
- Improve understanding of toxicity.
- Predict unbound drug concentration-time profiles in infants undergoing surgery.
- Through simulation methods, consider safety of alternative dosing schedules.

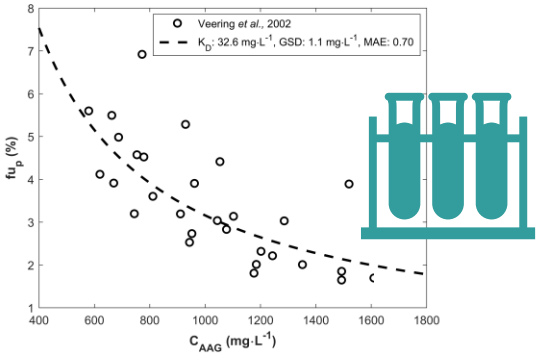
Vashisht *et al*. 2019

Nonlinear mixed-effects models
clinical study design and modelling

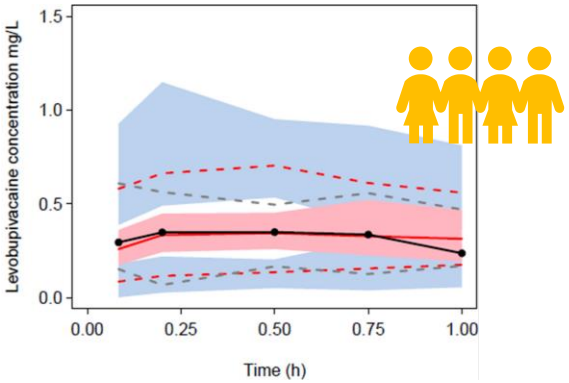


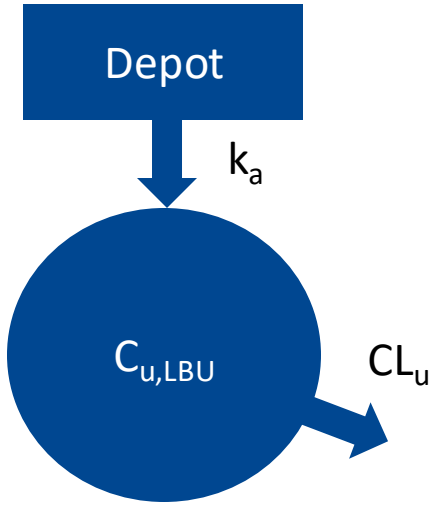
Darwich *et al*. 2017

Mechanistic model for prediction

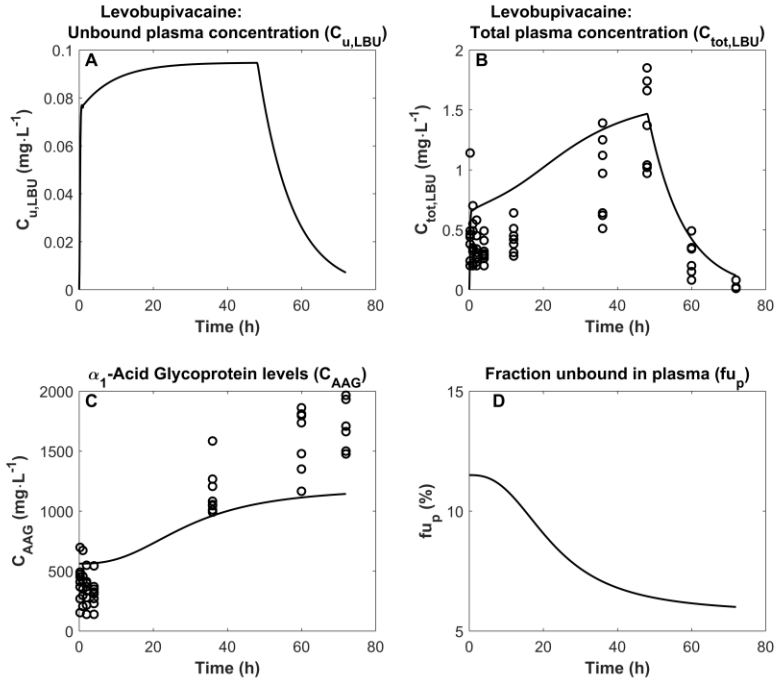




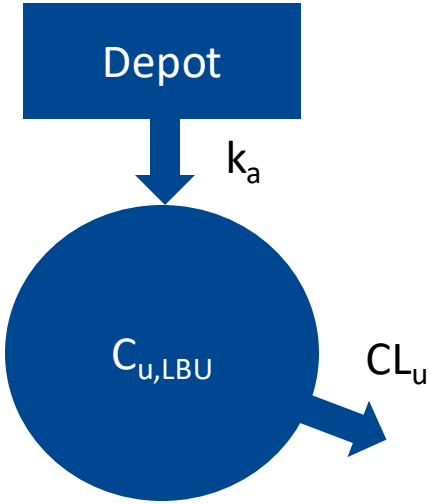




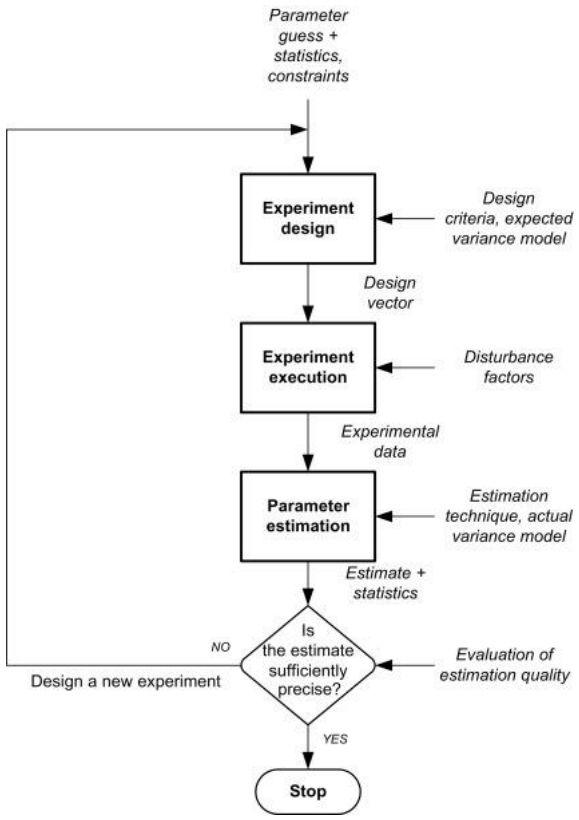
$$\frac{dA_{u,LBU}}{dt} = k_a \cdot A(t)_{LBU,depot} - A(t)_{u,LBU} \cdot \frac{CL_u}{V_u}$$

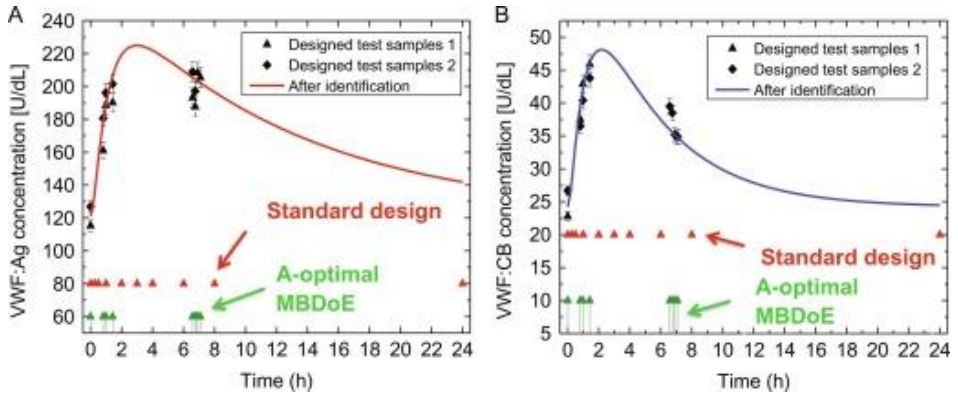


D-Optimal Study Design



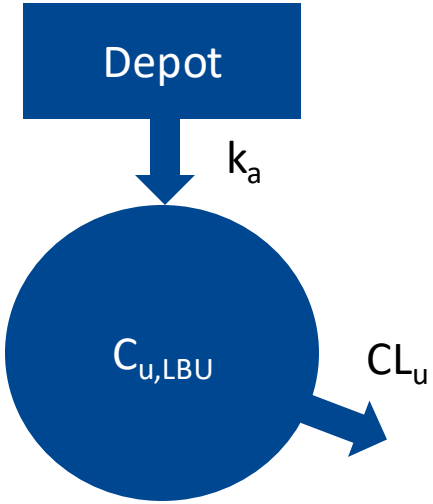
$$\frac{dA_{u,LBU}}{dt} = k_a \cdot A(t)_{LBU,depot} - A(t)_{u,LBU} \cdot \frac{CL_u}{V_u}$$



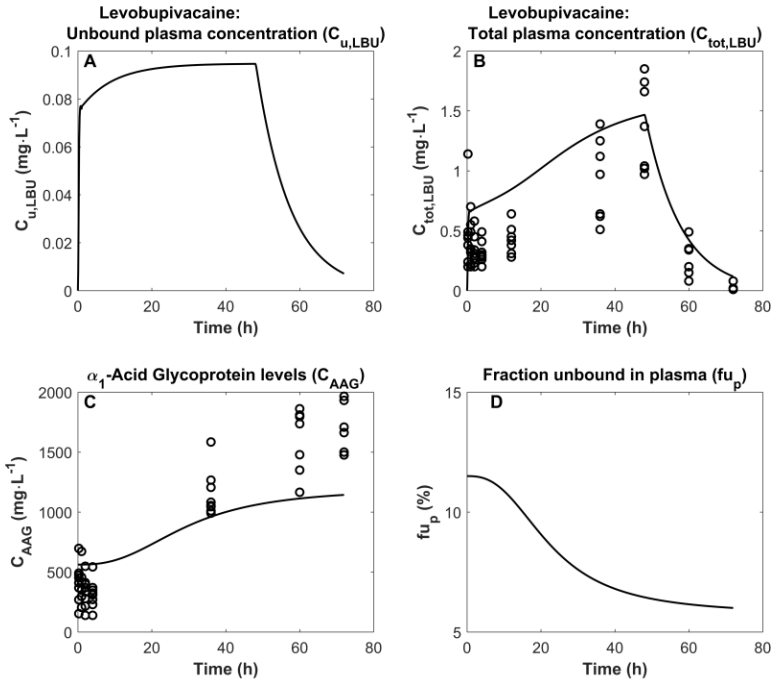


Galvanin and Bezzo 2018

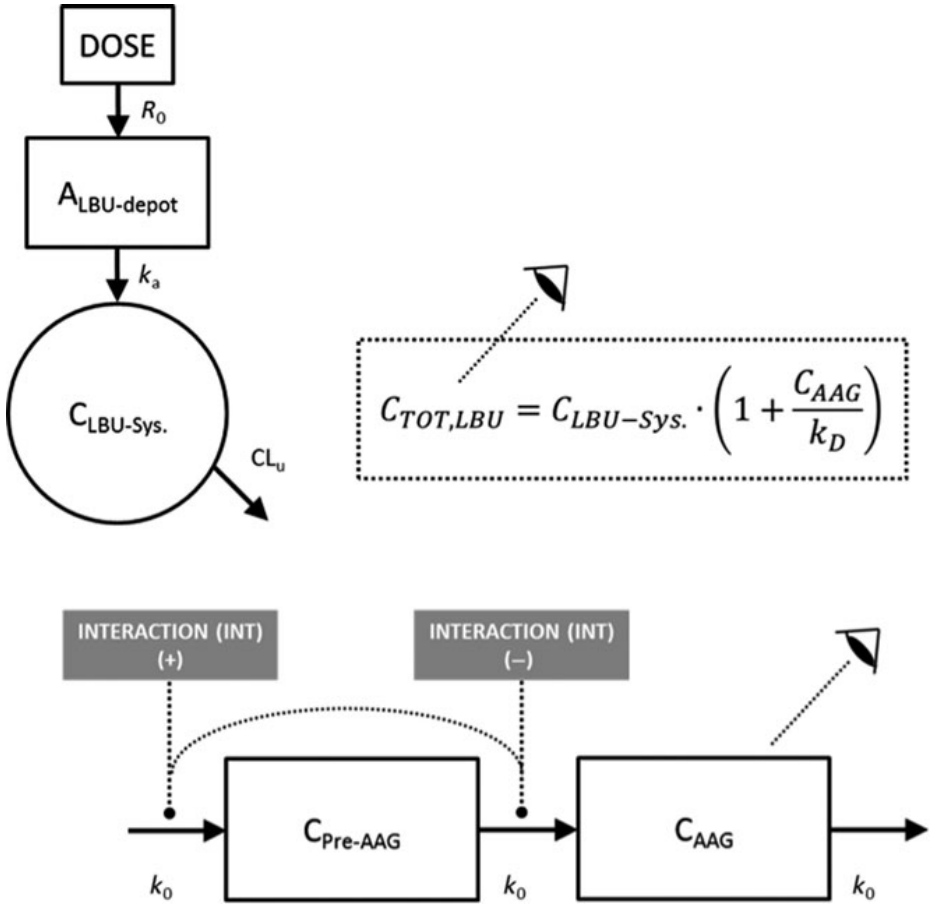
Mechanistic model for prediction

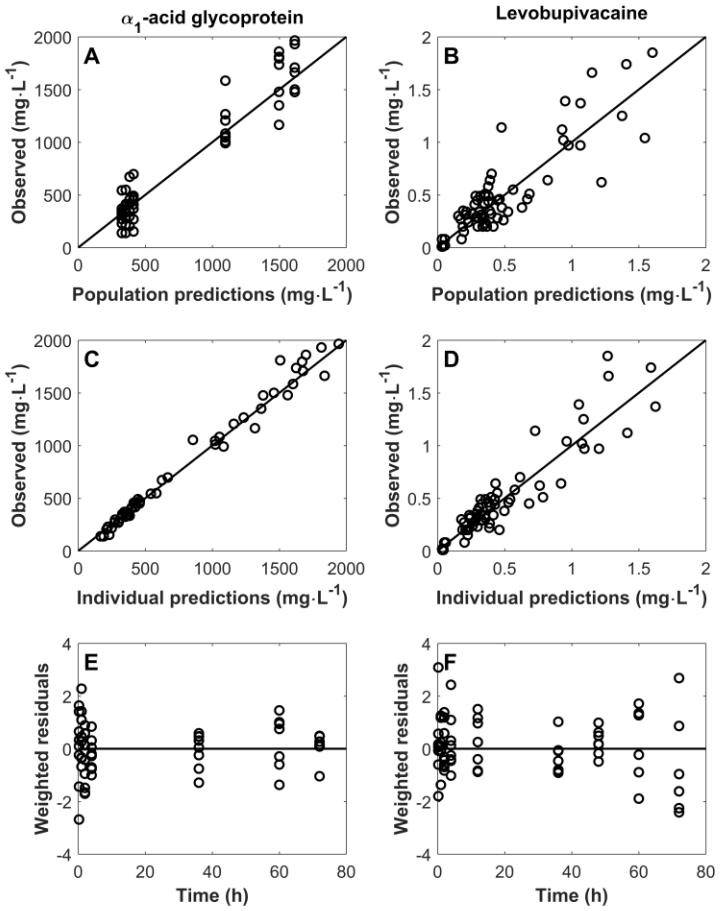


$$\frac{dA_{u,LBU}}{dt} = k_a \cdot A(t)_{LBU,depot} - A(t)_{u,LBU} \cdot \frac{CL_u}{V_u}$$



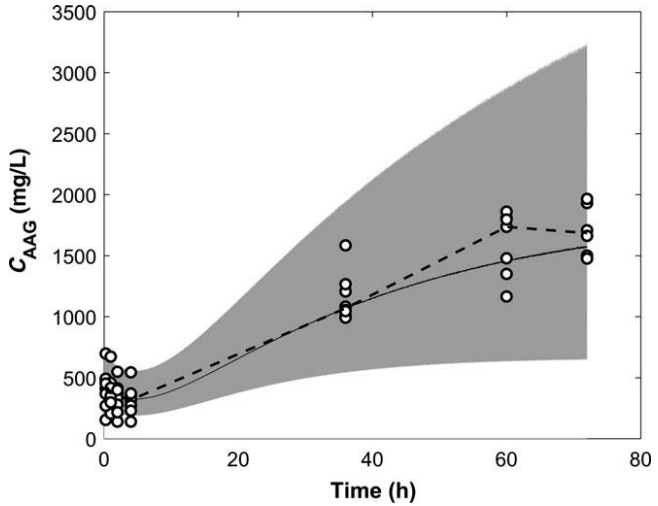
Nonlinear mixed-effects models
Modelling the clinical study data





Nonlinear mixed-effects models

Modelling the clinical study data



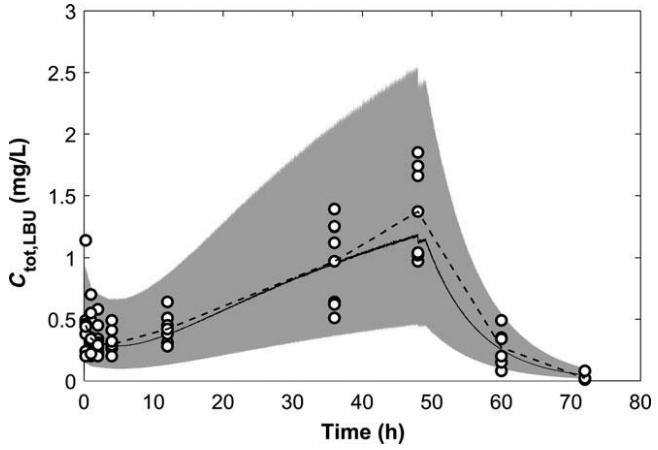


TABLE 1 α_1-acid glycoprotein (AAG) dynamic model and levobupivacaine pharmacokinetic model, final parameter estimates

Parameter	Estimate	%BSV	95%CI
AAG parameters			
CL_0 (L/h)	0.136	—	0.125, 0.141
C(0) (mg/L)	413	31.2	292, 475
INT	4.58	54.6	3.50, 5.13
Err prop (%)	7.81	—	4.25, 14.34
Levobupivacaine parameters			
k_a (/h)	0.15	—	0.138, 0.156
V_u/F (L)	1.05	—	0.62, 1.27
CL_u/F (L/h)	61.3	28.1	48.1, 68.0
K_D (mg/L)	32.6	—	Fixed
Err prop (%)	33.0	—	12.15, 89.72

BSV, interindividual variability presented as %CV (coefficient of variation); CL_u/F, apparent unbound clearance; CL_0, conversion clearance; C(0), concentration of AAG and pre-AAG at time zero; Err prop, proportional residual variability; INT, interaction term; k_a, first-order absorption rate; V_u/F, apparent unbound volume of distribution; 95% CI are the 95% confidence intervals.

Nonlinear mixed-effects models
Predictions and implications

